Computer Networks

UNIT-I: INTRODUCTION AND PHYSICAL LAYER

Introduction:

I. Network Hardware

II. Network Software

III. Reference Models - OSI, TCP/IP

IV. Example Networks – Internet; Wireless LANs - 802.11.

Physical Layer:

I. Guided Transmission Media

II. Wireless Transmission.

Introduction

- The Historical Perspective
  - The 18th century: the great mechanical systems accompanying the Industrial Revolution.
  - The 19th century: steam engine.
  - The 20th century: information gathering, processing, and distribution.
  - The 21th century: Internet, large distributed systems (e.g., Grid), heavy reliance on computers.

Introduction (Cont'd)

- What is Computer Network?
  - we will use the term "**computer network**" to mean a **collection of autonomous computers interconnected by a single technology.**
  - Two computers are said to be interconnected if they are able to exchange information.
  - copper wire; fiber optics, microwaves, infrared, and communication satellites, etc.

Purpose: To share the resources

- Sharing the Files/ Documents

- Sharing the Printer

- Sharing the server

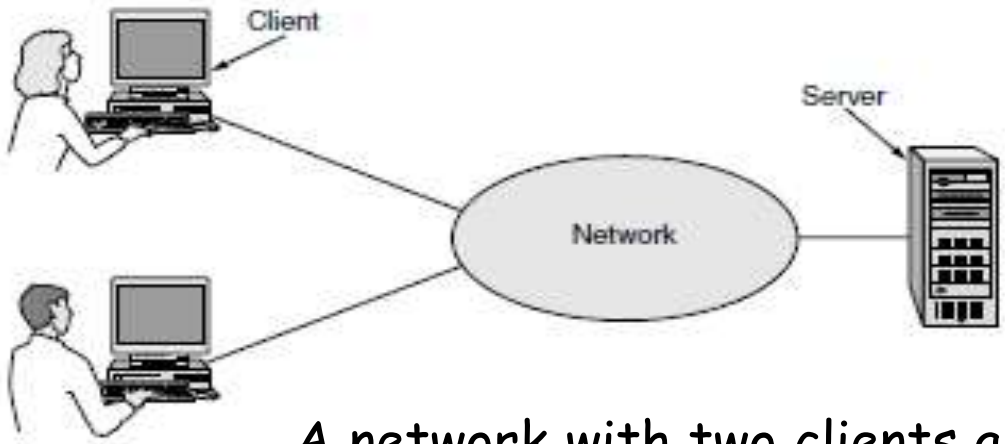
Types	Architecture	Devices	Medium	Protocols
LAN	Peer-to-Peer	Modem	Coaxial Cable	TCP/IP
MAN	Client server	Router	Twisted pair	UDP
WAN	OSI Layer	Bridge	Wireless	
	TCP/IP Layer	Switch		

Topology	Technology
Bus	Internet
Star	Intranet
Ring	Extranet

USE OF COMPUTER NETWORKS

1. Business applications.
2. Home Applications.
3. Mobile Users .
4. Social Issues.

1. Business applications- Resource sharing

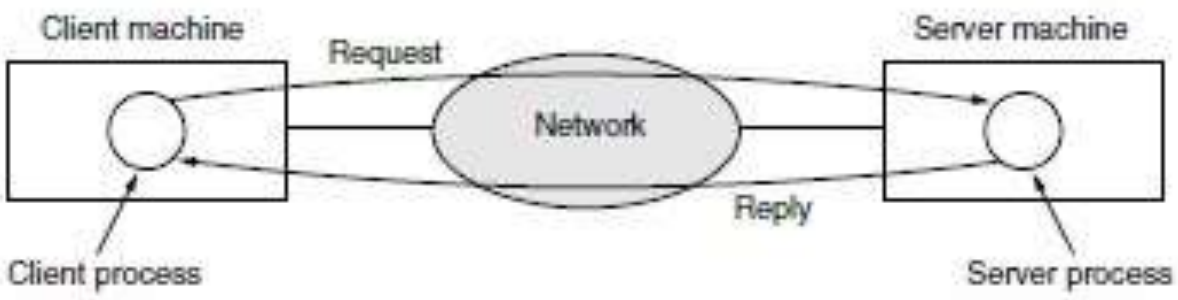


A network with two clients and one server.

Computer Network Goals:

1. Client-server model- Web application.
2. A second goal of a computer network can provide a powerful **communication medium among employees**.- **email (electronic mail).** Telephone calls between employees may be carried by the computer network instead of by the phone company. This technology is called **IP telephony or Voice over IP (VoIP).**
3. A third goal for many companies is doing business electronically, especially with customers and suppliers. This new model is called **e-commerce (electronic commerce)**

This reduces the need for large inventories and enhances efficiency



The client-server model involves requests and replies.

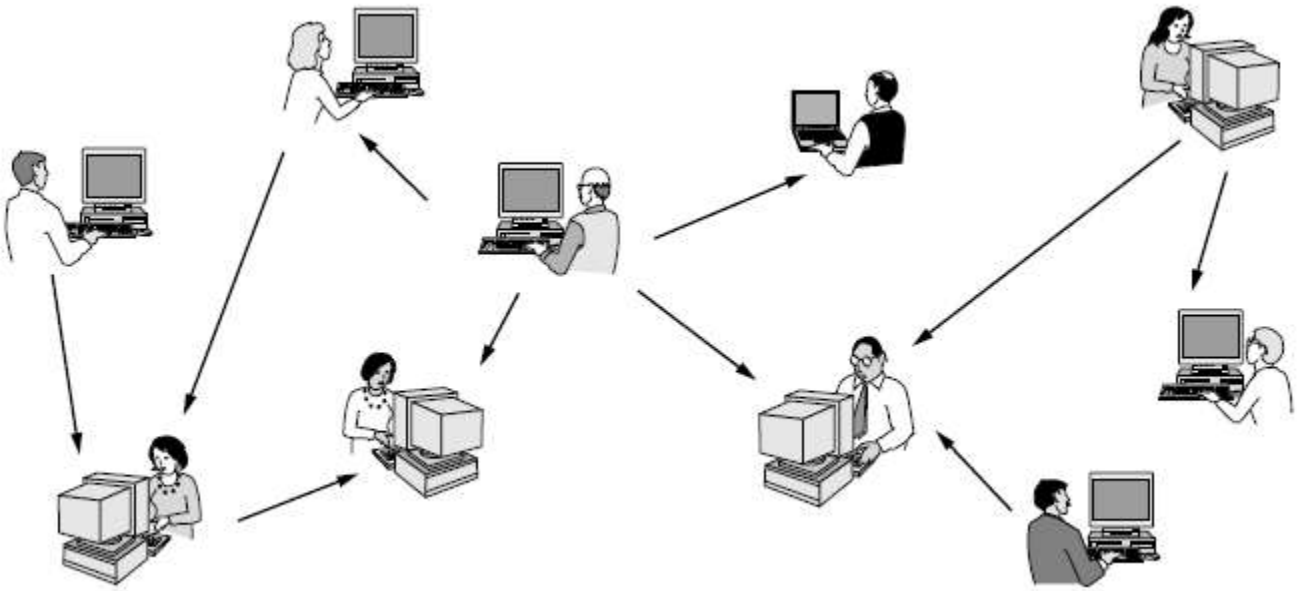
2. Home Applications

•Internet access provides home users with **connectivity to remote computers.**

The more popular uses of the internet for home users are as follows:

1. Access to remote information.
2. Person-to-Person Communication.- **peer-to-peer communication**
3. Interactive Environment.
4. Electronic Commerce.

peer-to-peer communication:



In a peer-to-peer system there are no fixed clients and servers

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books online
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products online
P2P	Peer-to-peer	Music sharing

Some forms of e-commerce

3. Mobile User

Wireless	Mobile	Typical applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in unwired buildings
Yes	Yes	Store inventory with a handheld computer

Combinations of wireless networks and mobile computing

PDAs' – Personal Digital Assistant
WAP 1.0 Wireless Application Protocol.
WAP 2.0
Mobile Commerce- M-commerce

4. Social Issues

Gamil, cookies

I. Network Hardware

2 types of transmission technologies

- Broadcast
  - Broadcast networks have a single communication channel that is shared by all the machines on the network.
  - Short messages, called packets in certain contexts, sent by any machine are received by all the others. (**Address Checking required**)

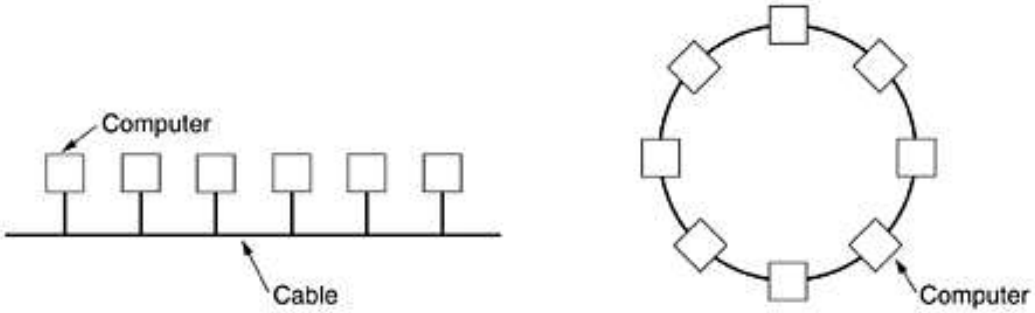
- Point-to-point
  - In point-to-point networks, there consist of many connections between individual pairs of machines.
- As a general rule (although there are many exceptions), smaller, geographically localized networks tend to use broadcasting, whereas larger networks usually are point-to-point.

- Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called unicasting.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

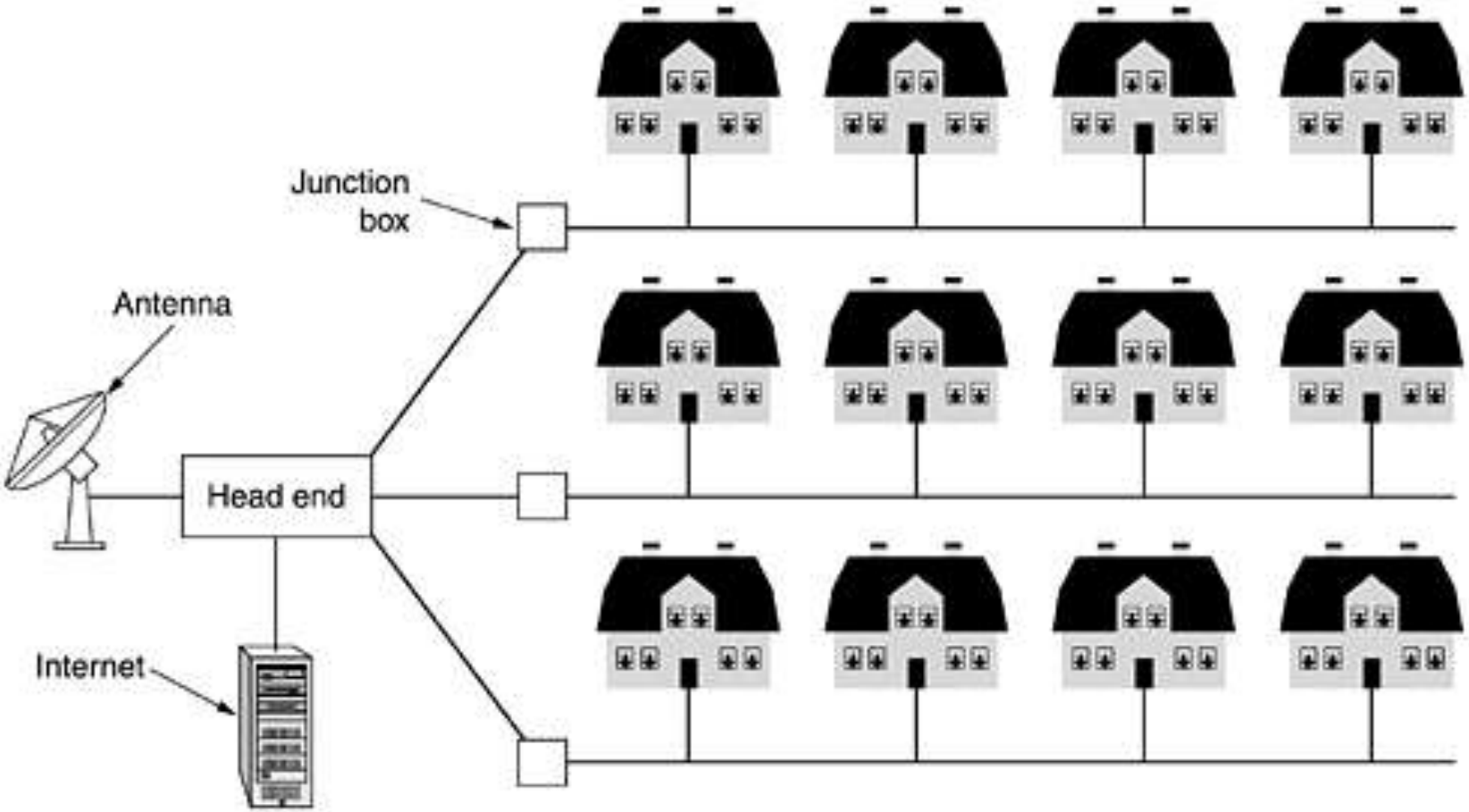
Classification of interconnected processors by scale

- **LAN: local area network**
  - Local area networks, generally called LANs, are privately-owned networks within a single building or campus of up to a few kilometers in size.
  - LANs may use a transmission technology consisting of a **cable** to which all the machines are attached.
  - Traditional LANs run at speeds of 10 Mbps to 1000 Mbps, have low delay (microseconds or nanoseconds), and make very **few errors**.
  - Various topologies are possible for broadcast LANs.
  - **Lack of Privacy-** accessed by all



(a) Bus (b) Ring

MAN: metropolitan area network



MAN:

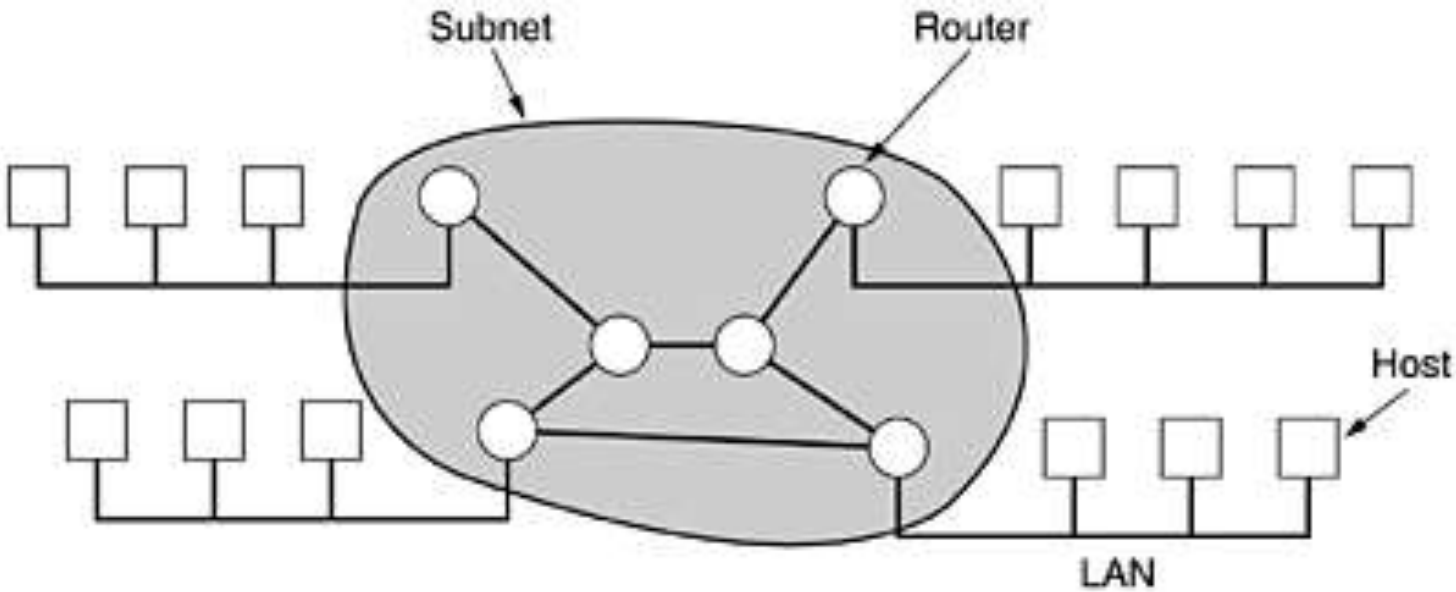
- Connection of Different LANs
- With in the city – ex: Bank
- Connecting all the branches in a city
- Sharing of Data
- Less Security
-Fiber Optics or cables used- Requires more cable

WAN:

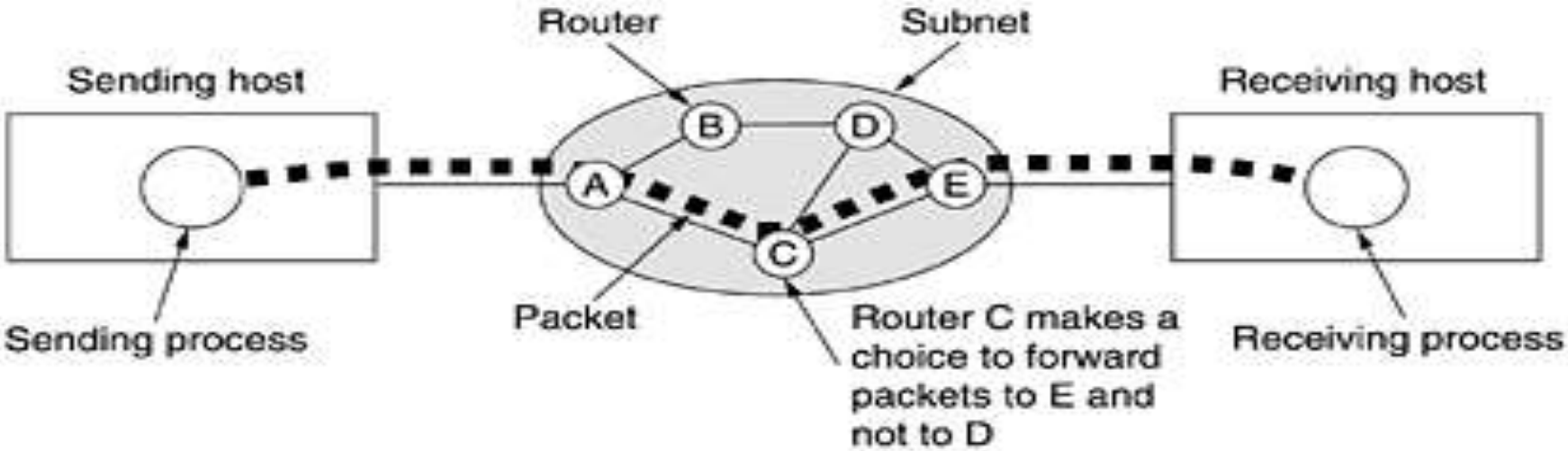
- Connection will be among different cities (MANs)
- **Transmission medium-** Satellite (or) Telephone cable
- Data sharing using Routers
- Less Security
- Need to have a Firewall- Ex: Internet

- **WAN and Internet**
  - Hosts, **The hosts are owned by the** customers (e.g., people's personal computers).
  - Subnet, **typically owned and operated by a** telephone company or Internet service provider (ISP).
  - **The hosts are connected by subnets**.
  - the **subnet consists of two distinct components:** **transmission lines** and **switching elements**. Transmission lines move bits between machines. Switching elements (routers) are specialized computers that connect three or more transmission lines.

WAN and Internet

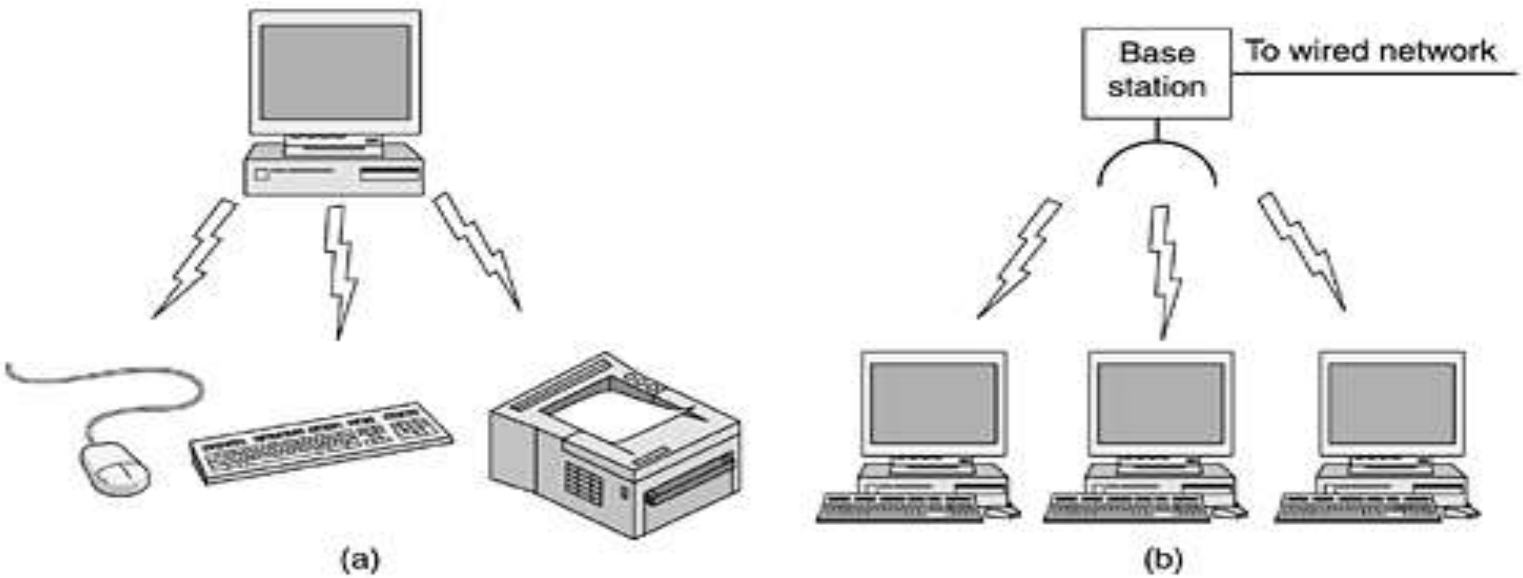


Simple WAN (Internet) illustrated



Packets routing

- Wireless Networks
  - Bluetooth: short range wireless network (<10m).
  - Wireless LANs: for moderate range, becomes more and more common today (<100m).
  - network used for cellular telephones: distances involved are much greater and the bit rates much lower (above 1km).



Network Topologies

Topology:

Arranging computers in a Network

Types:

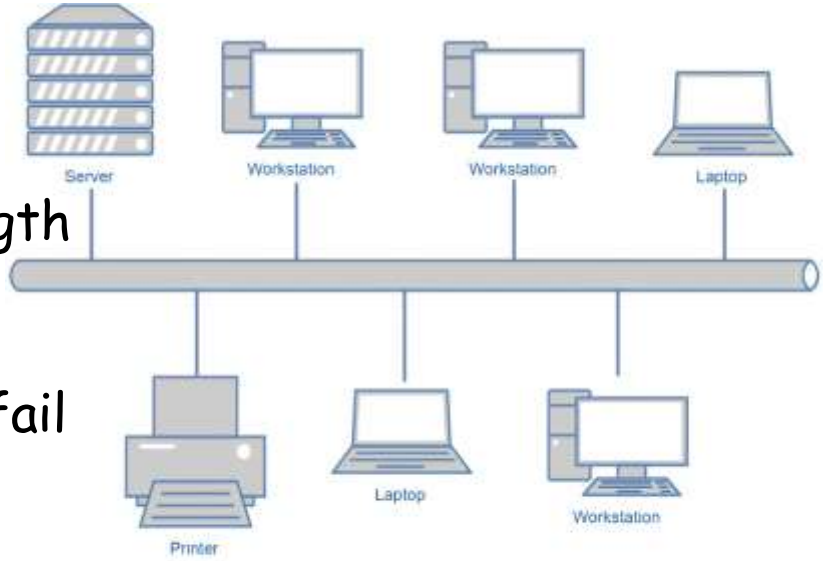
- Bus Topology
- Star Topology
- Ring Topology
- Mesh Topology
- Hybrid Topology

Bus Topology:

-Single cable
- Due to header the message will be received by the correct one

Drawbacks:
- Collision
- More no.of computers – Signal strength will be low
- No security
- If cable fails complete network will fail

Advantages:
- Less Expensive
- Easy of Configure
- Installation is easy



Bus Topology Network

Star Topology:

- Centralized device- Hub/ Switch
- Hub:

Transmission- Broadcast

Ex: PC1 --- PC4

PC1-Hub- All PCs connected to Hub

- Switch:

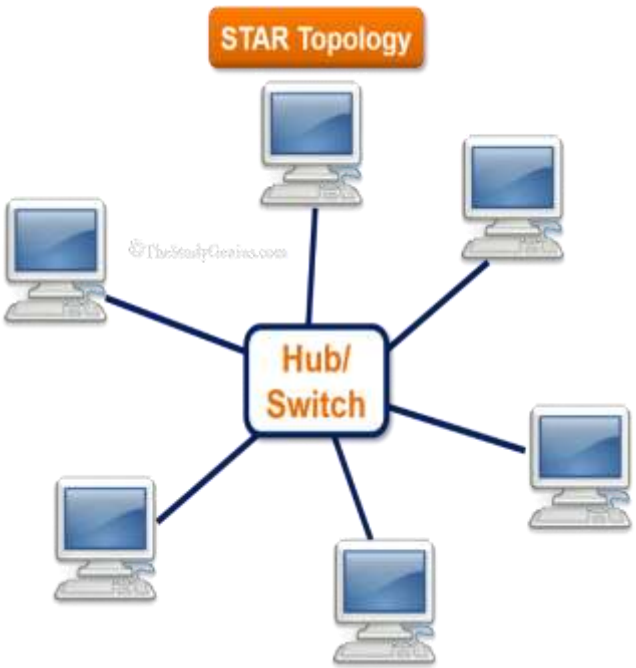
Transmission- Unicast

Ex: PC1---PC4

PC1-Switch-PC4

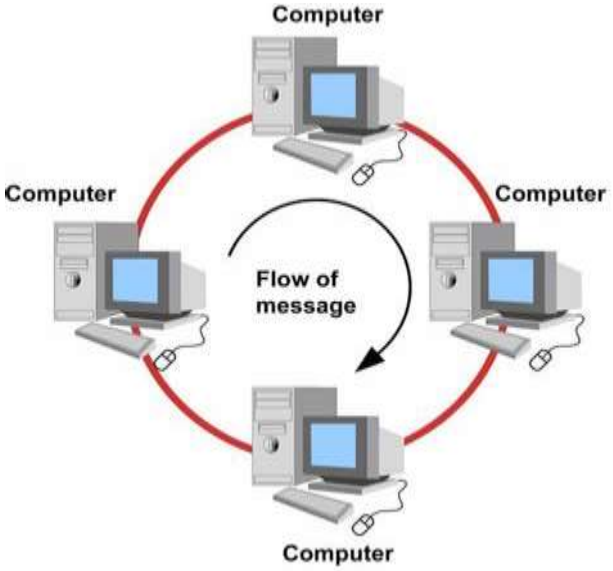
Adv: Security– Switch, Less expensive

Dis Adv: centralized device Fails--- Network Will Fail

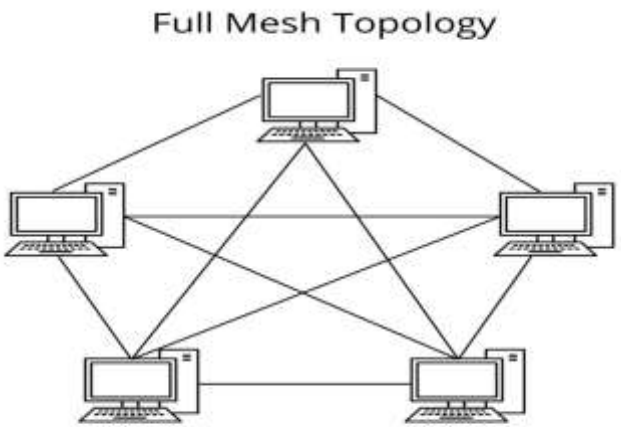


Ring Topology:

- No Collisions
- Expensive
- Difficult to Reconfigure
- If System fails, Transmission fails.
- Easy to fault tolerance
- Less Security



Mesh Topology:



- n – nodes
- Links for each node- (n-1),
- Total links- n(n-1)/2 – For bidirectional
- If Unidirectional- total links are n(n-1).
- Drawbacks:

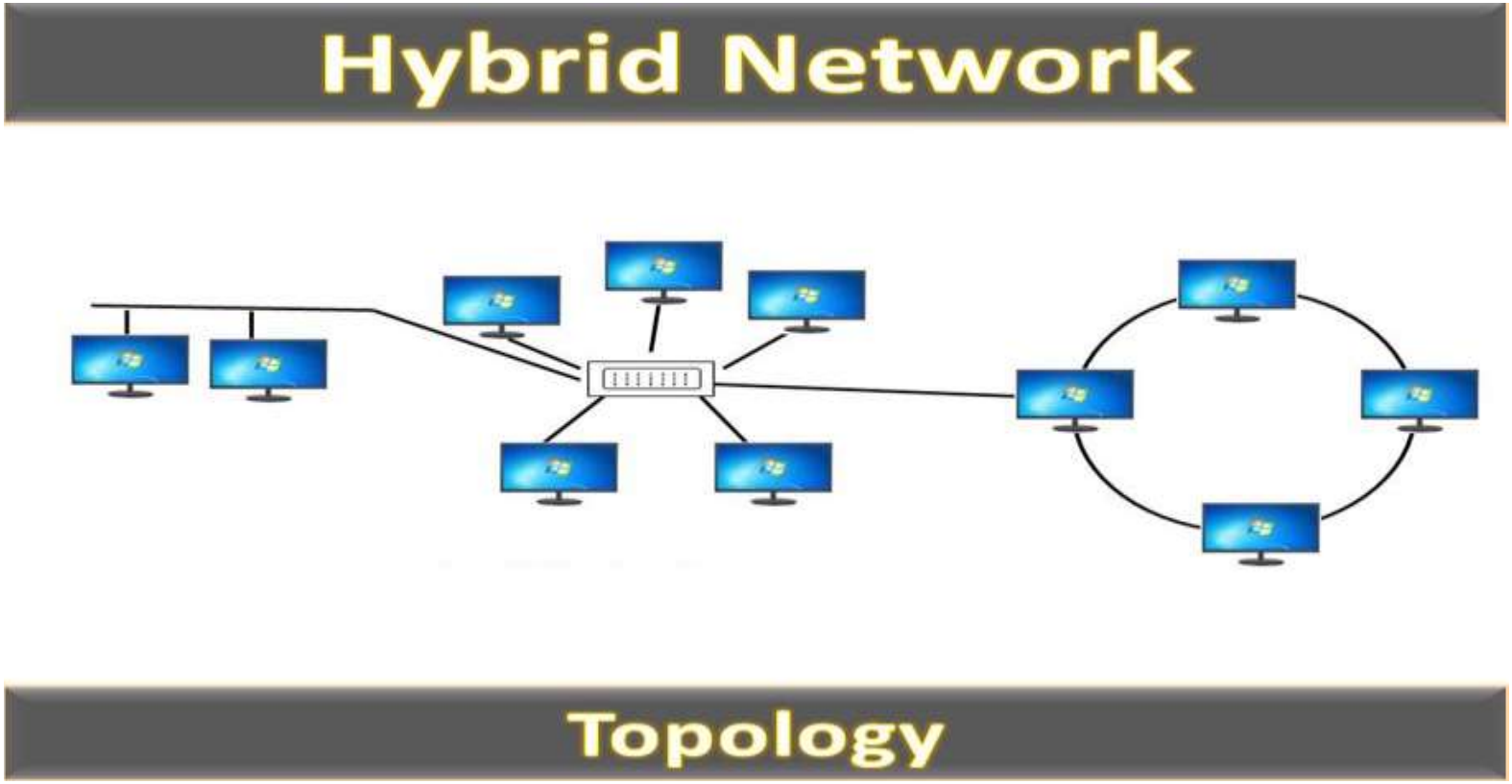
Expensive

Difficult to reconfigure

If system fails transmission will not fail (There will be some alternate)

Hybrid Topology:

- Connection of multiple topologies.



Network Devices

- Repeater: Used to regenerate the signal (To maintain signal strength- Lost the message, or corrupted – need to resend)- Physical Layer.
- Hub: Used to transmit data packets to all connected devices.
- Bridge: Interconnecting two LANs working on same protocol – Data Link layer
- Switch: Error checking will be done. Good packets will send to correct ports. Data packets with errors will not be forwarded.
- Router: Packets will routed to destination in best routing path. Dynamically routing will be done – Network Layer.
- Gateway: Passing the data to the networks of different networking models. – Protocol Converter.

II. Network Software

Network Software

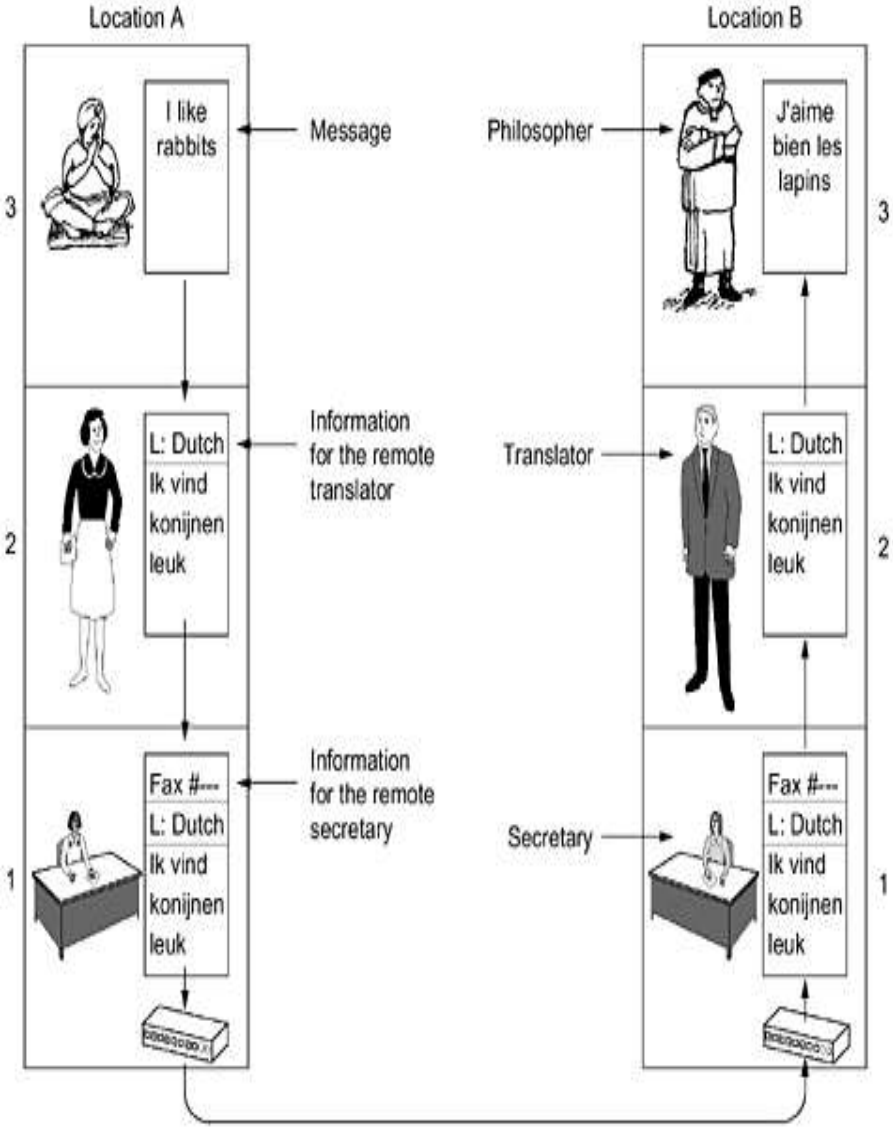
1. Protocol Hierarchies
2. Design Issues for the Layers
3. Connection-Oriented and Connectionless Services
4. Service Primitives
5. The Relationship of Services to Protocols

Network Software (Cont'd)

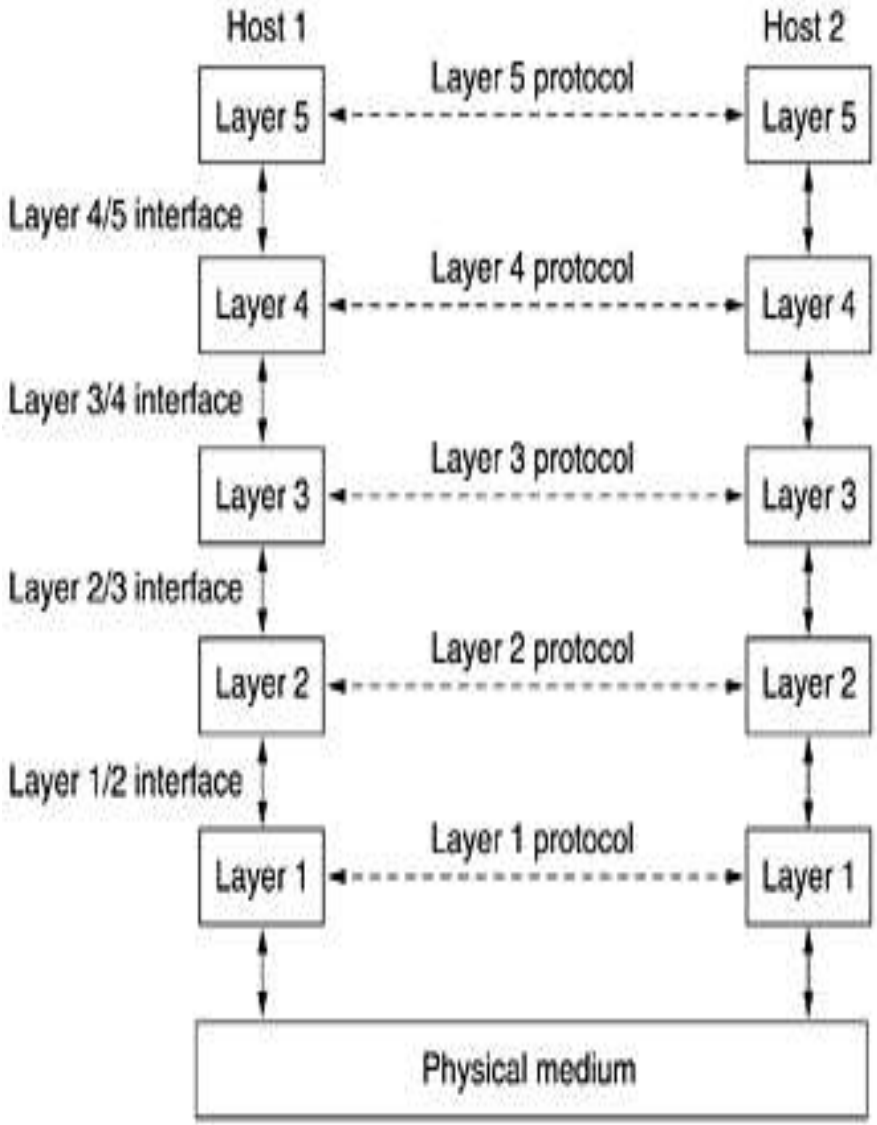
1. Protocol Hierarchies

- To reduce their design complexity, most networks are organized as a **stack** of **layers** or **levels**, each one built upon the one below it.
- The purpose of each layer is to offer certain services to the higher layers, shielding those layers from the details of how the offered services are actually implemented.
- A **protocol** is an agreement between the communicating parties on how communication is to proceed.

Network Software(Cont'd)



The philosophers analogy



The layered computer network

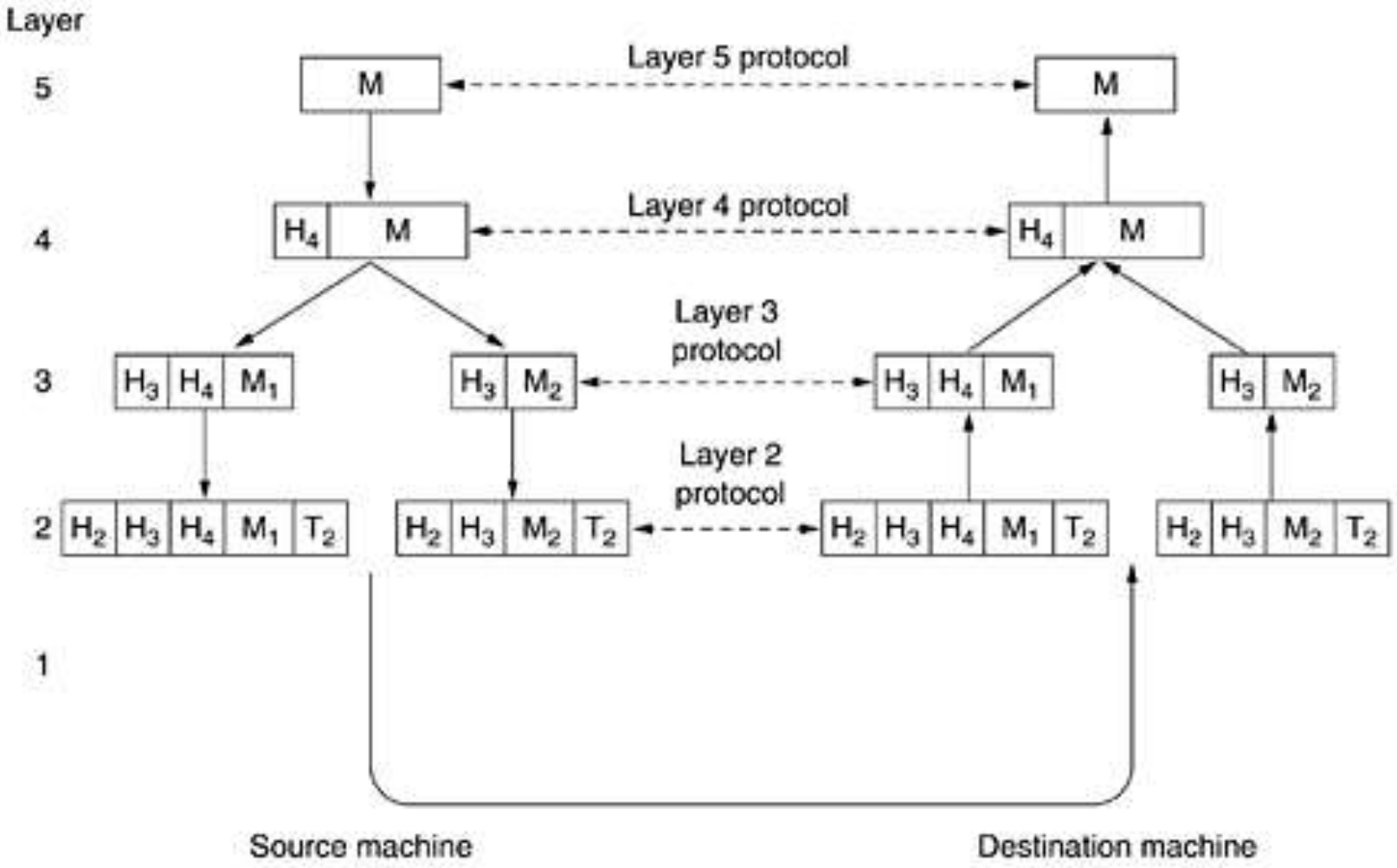
Network Software (Cont'd)

- The key terms
  - Layers; Protocol;
  - Interface: Between each pair of adjacent layers is an interface.
  - Network architecture: A set of layers and protocols is called a network architecture.
  - Protocol stack: A list of protocols used by a certain system, one protocol per layer, is called a protocol stack.

Network Software (Cont'd)

An example network protocol stack

M- Message, H- Header



Network Software (Cont'd)

- Message is generated by the application of the source machine.
- Message will be sent from the source to the destination.
- Message "M" is transferred from layer 5 to layer 4, with a header containing control information, such as sequence numbers, which helps layer 4 maintain the message order.
- Layer 3 break the message from layer 4 into two pieces to fit the transmission restrictions, while adding another header to tell layer 2 where the dest. is.
- Layer 2 adds the messages from layer 3 with another header, telling the actual (physical) address of the dest, and a trailer, which is the checksum of the message for correction assertion.
- At the receiving machine the message moves upward, from layer to layer, with headers being **stripped off** as it progresses. Message is sent to the dest. machine.

Network Software (Cont'd)

2. Design Issues for the Layers

1. How is it possible that we find and fix these errors?

- One mechanism for finding errors in received information uses codes for error detection & correction.
- Error control is an important issue because physical communication circuits are not perfect.

Another reliability issue is finding a working path through a network. Often there are multiple paths between a source and destination, and in a large network, there may be some links or routers that are broken.- Routing

2. A second design issue concerns the evolution of the network.

- **protocol layering-** Every layer needs a mechanism for identifying senders and receivers. This mechanism is called addressing or naming, in the low and high layers, respectively.
- An aspect of growth is that different network technologies often have different limitations.- mechanisms for disassembling, transmitting, and then reassembling messages. This overall topic is called internetworking.
- When networks get large, new problems arise. Cities can have traffic jams, a shortage of telephone numbers, and it is easy to get lost.- **scalable.**

3. A third design issue is resource allocation.

- Design- statistical multiplexing, meaning sharing based on the statistics of demand.
- An allocation problem that occurs at every level is how to keep a fast sender from swamping a slow receiver with data.- **flow control.**

- Too many computers want to send too much traffic, and the network cannot deliver it all. This overloading of the network is called **congestion.**

4.The last major design issue is to secure the network by defending it against different kinds of threats.

-provide **confidentiality** defend against this threat, and they are used in multiple layers.

-Mechanisms for **authentication** prevent someone from impersonating someone else.

- mechanisms for **integrity** prevent surreptitious changes to messages, such as altering "debit my account $10" to "debit my account $1000."

-All of these designs are based on **cryptography**

Summery of Design Issues:

- Addressing or Naming
- Error Control
- Flow Control --- Congestion
- Statistical Multiplexing
- Routing
- Scalable- ex: Population increases- Maintain the scalability
- QoS– Quality of Service– Real time
- Reliability
- Security

Network Software (Cont'd)

3. Connection-Oriented and Connectionless Services?

- Connection-Oriented Service: the service user first establishes a connection, uses the connection, and then releases the connection. (e.g., the telephone)
- Connectionless Service: Each message carries the full destination address, and each one is routed through the system independent of all the others. (e.g., the postal system) Usually, connectionless service can not guarantee the order of messages.
- In order to enhance the reliability of transmission of connection-oriented service, acknowledge each received message is helpful. For example, the file transfer.
- However, some applications prefer fast speed than the reliability. For example, the digitized voice traffic, video conference.

Reliable connection-oriented service has two minor variations:

1. Message Sequences - When two 1024-byte messages are sent, they arrive as two distinct 1024- byte messages, never as one 2048-byte message.
2. Byte Streams- the connection is simply a stream of bytes, with no message boundaries.

   To download a **DVD movie, a byte stream** from the server to the user's computer is all that is needed.

   For some applications, the transit delays introduced by acknowledgements are unacceptable.

   One such application is digitized voice traffic for voice over IP.

Unreliable (meaning not acknowledged) connectionless service is often called datagram service, in analogy with telegram service, which also does not return an acknowledgement to the sender.

- Six different types of service

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Remote login
	Unreliable connection	Digitized voice
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Registered mail
	Request-reply	Database query

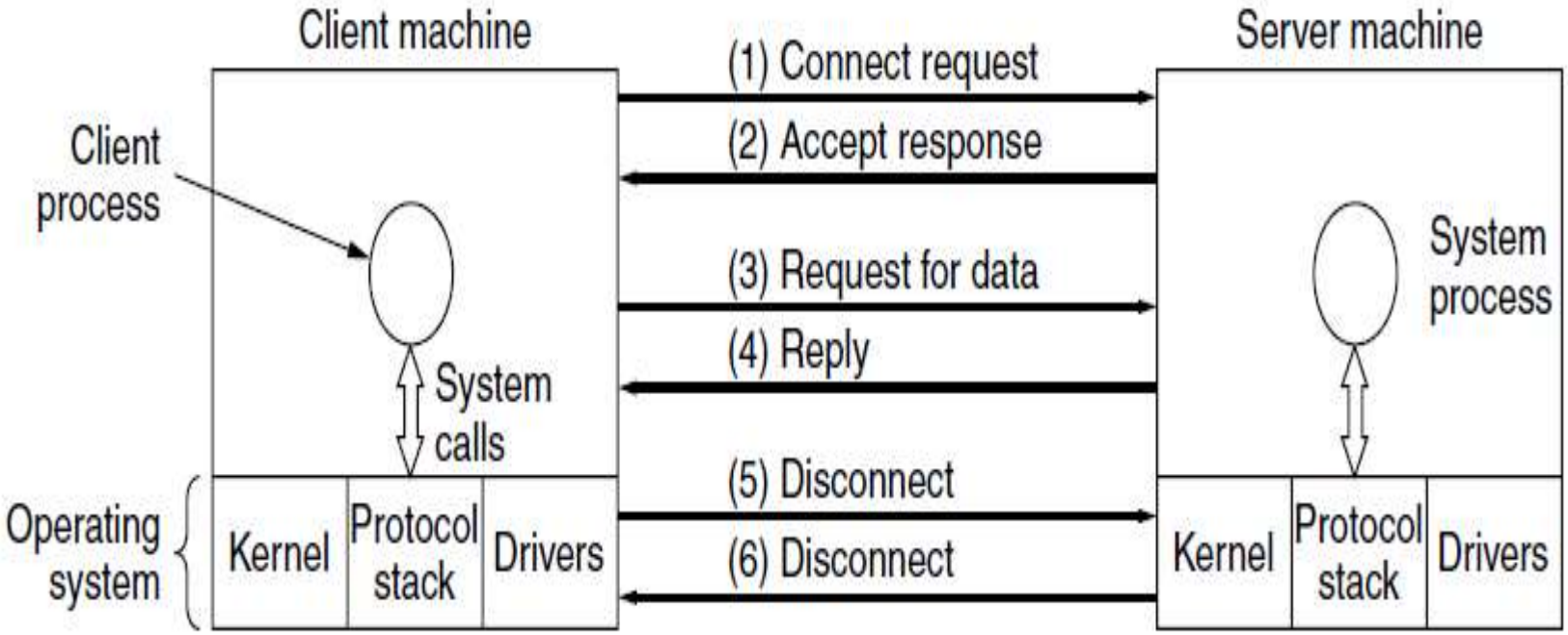
both reliable and unreliable connection-oriented and connectionless communication coexist

4. Service Primitives

- A service is formally specified by a set of primitives (operations) available to user processes to access the service.
- These primitives tell the service to perform some action or report on an action taken by a peer entity.
- The set of primitives available depends on the nature of the service being provided.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Six service primitives that provide a simple connection-oriented service.

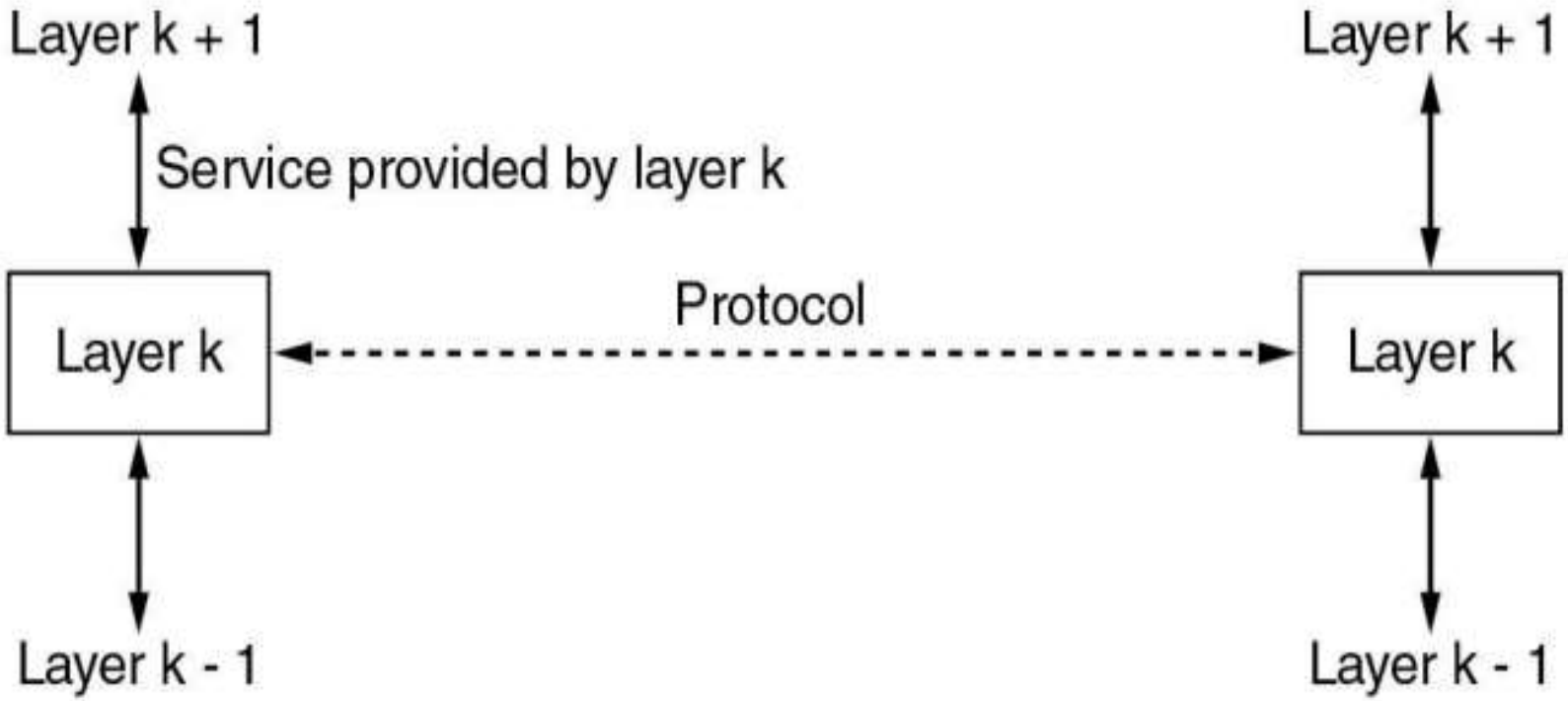


- simple client-server interaction using acknowledged datagrams

- Packets sent in a simple client-server interaction on a connection-oriented network

Services to Protocols Relationship

- The relationship between a service and a protocol.



III. Reference Models

Two important network architectures:

- The OSI reference model (Open Systems Interconnection)
- The TCP/IP reference model (Transmission Control Protocol / Internet Protocol).

OSI reference model: (Day and Zimmermann, 1983)

- Developed by the International Standards Organization (ISO) so it is called as ISO OSI reference model because it deals with connecting open systems for communication with other systems.
- It is rarely used today.
- OSI model itself is **not a network architecture** because it does not specify the exact services and protocols to be used in each layer.
- Although the protocols associated with the OSI model are not used any more, the model itself is actually quite general and still valid, and the features discussed at each layer are still very important.

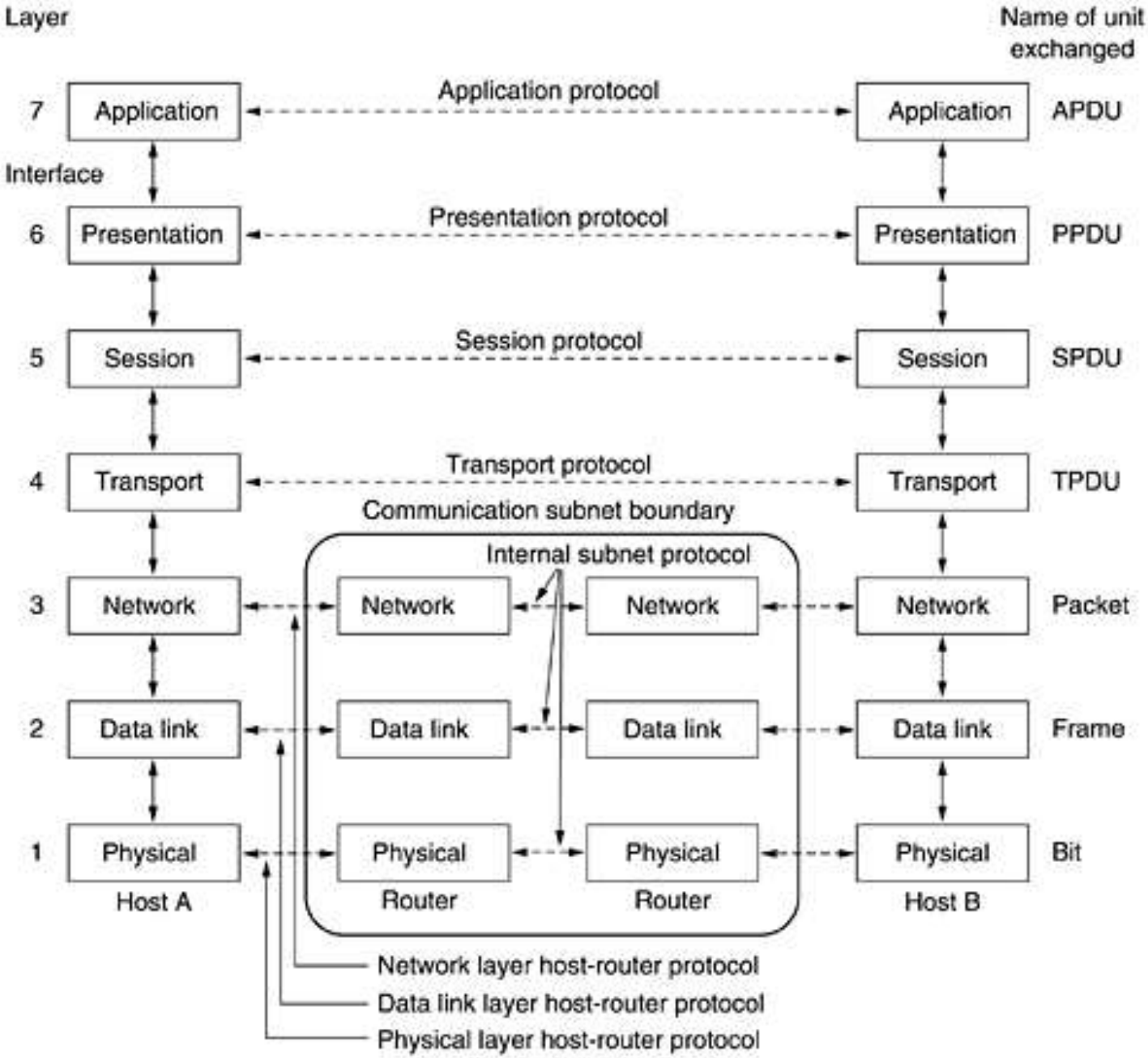
TCP/IP reference model: developed by ARPANET (Advanced Research Projects Agency Network)- Used to construct the Internet today.

- The model itself is not of much use but the protocols are widely used.

The OSI model has seven layers.

The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.



OSI Reference model

1. The Physical Layer

- The physical layer is concerned with transmitting raw bits over a communication channel.
- Voltage representation: 0 volts for 0 bit and 5 volts for 1 bit.
- Physical media: Wired & Wireless- Twisted pair cables, coaxial cables, fibre optics & Radio , infrared, blue tooth.
- Topology: Physical Structure of N/W.
- Transmission mode: Simplex, half duplex, full duplex.
- Transmission technology: Broadcast & point-to-point.
- Bit rate (no. of bits /sec) & Baud rate (no. of bits successfully transmitted /sec) .
- Design issues largely deal with mechanical, electrical, and timing interfaces, as well as the physical transmission medium, which lies below the physical layer.

2. The Data Link Layer

- The main task of the data link layer is Framing (data frames- typically a few hundred or a few thousand bytes) and transmit the frames sequentially.
- If the service is reliable, the receiver confirms correct receipt of each frame by sending back an acknowledgement frame.
- Physical Address- Name, Roll no.
- Flow (bits checking) & Error (detection & correction) Control.
- Acks (acknowledgement) : +ve- data success- ACK
  & -ve- data loss- NAK.
- Data retransmission (fail or retransmit) and Error frames (damage or loss).
- Sequence no.: Access controlling & Accounting & billing.
  - having the sender break up the input data into data frames and transmit the frames sequentially.
- How to control access to the shared channel. A special sublayer of the data link layer, the medium access control sublayer, deals with this problem.

3. The Network Layer:

- The network layer controls the operation of the subnet(routing).
- A key design issue is determining how packets are routed from source to destination.

Network Layer issue: Handling congestion.

- The quality of service provided (delay, transit time, jitter, etc.)

4. The Transport Layer:

- The basic function of the transport layer is to accept data from above, **split it up into smaller units** if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end.
- The transport layer Services: The most popular type of transport connection is an error-free point-to-point channel that delivers messages or bytes in the order in which they were sent.
- The transport layer is a true end-to-end layer, all the way from the source to the destination.
- The difference between layers 1 through 3, which are chained, and layers 4 through 7, which are end-to-end.

5. The Session Layer

- The session layer allows users on different machines to establish sessions between them.
- Sessions offer various services, including Dialog Control (keeping track of whose turn it is to transmit), Token Management (preventing two parties from attempting the same critical operation simultaneously), and Synchronization (check pointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery).

6. The Presentation Layer

- concerned with the syntax and semantics of the information transmitted, to make it possible for computers with different data representations to communicate.

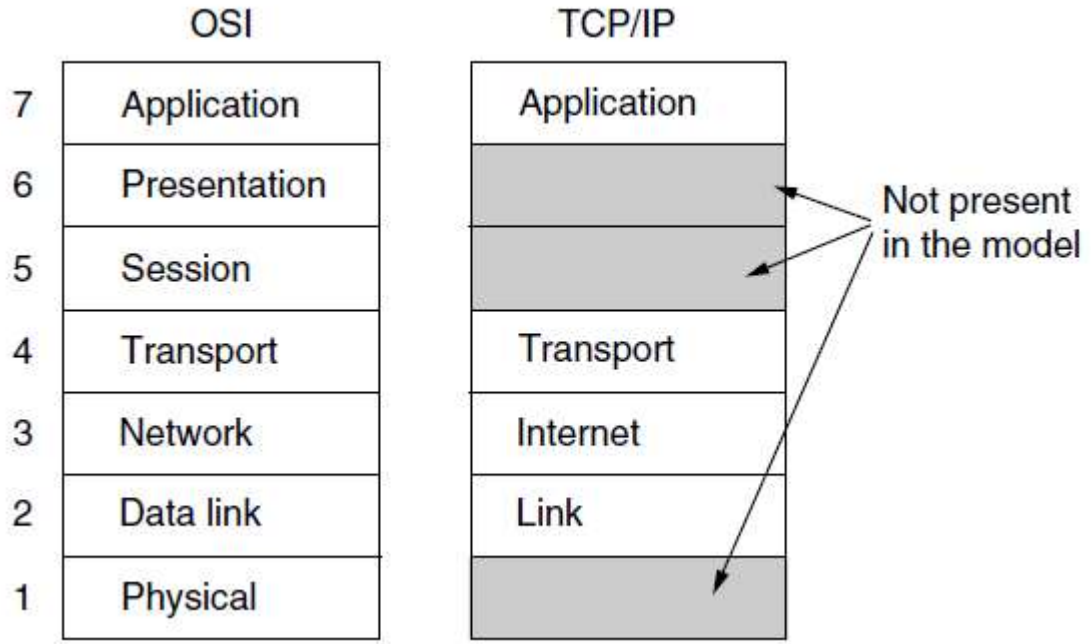
7. The Application Layer

- The application layer contains a variety of protocols that are commonly needed by users.
- One widely used application protocol is HTTP (Hyper Text Transfer Protocol), which is the basis for the World Wide Web.
- File Transfer, Electronic Mail, and Network News etc…

The TCP/IP Reference Model

The ARPANET was a research network sponsored by the DoD (U.S. Department of Defense).

It was first described by Cerf and Kahn (1974), and later refined and defined as a standard in the Internet community (Braden, 1989).
The design philosophy behind the model is discussed by Clark (1988).



The TCP/IP reference model

The Link Layer: describes what links such as serial lines and classic Ethernet must do to meet the needs of this connectionless internet layer. - an interface between hosts and transmission links.

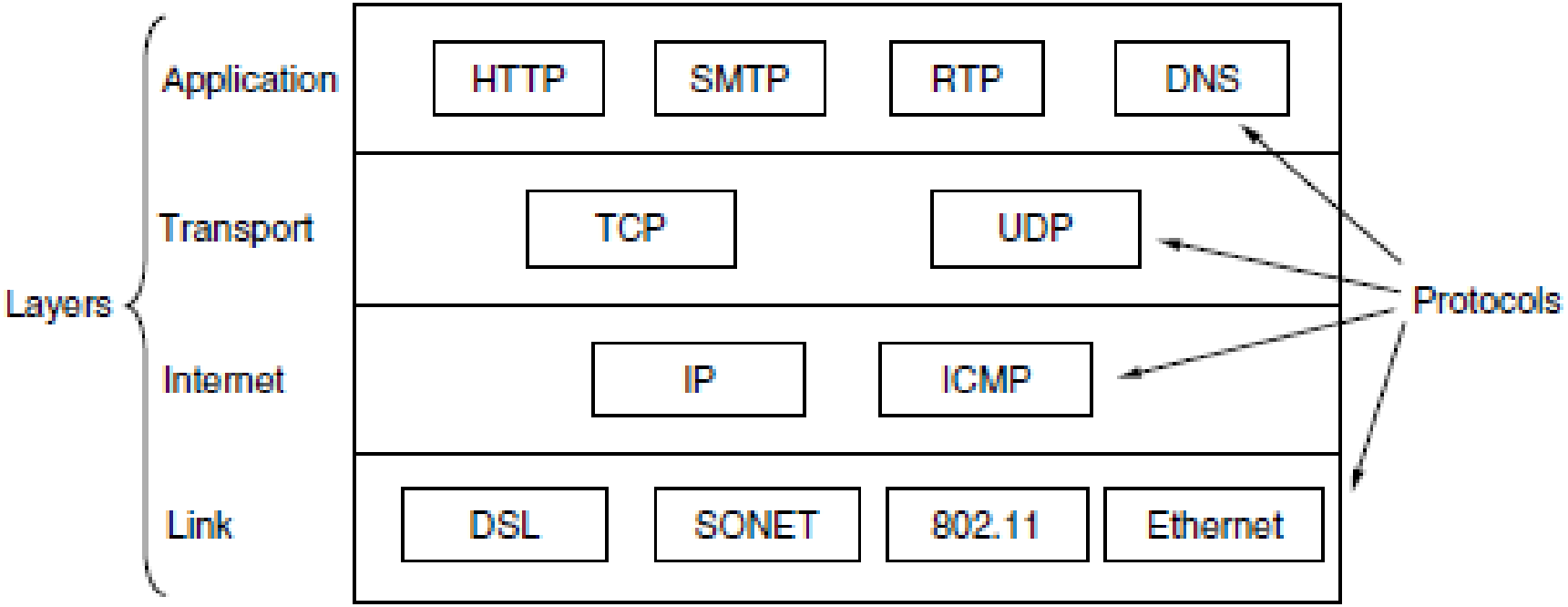
- **The Internet Layer :**
- Its job is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network).
- They may even arrive in a different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired.
- The internet layer defines an official packet format and protocol called IP (Internet Protocol).
- The job of the internet layer is to deliver IP packets where they are supposed to go.
- The underlining Host-to-Network Layer is a great void. host connected to this layer can send IP packets.

Introduction (Cont'd)

- The Transport Layer
  - It is designed to allow peer entities on the source and destination hosts to carry on a conversation
  - Two end-to-end transport protocols have been defined here. The first one, TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet.
  - The second protocol in this layer, UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own.

- **The Application Layer**
  - The TCP/IP model does not have session or presentation layers, which are of little use to most applications.
  - It contains all the higher-level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP).
  - Many other protocols have been added to these over the years: the Domain Name System (DNS) for mapping host names onto their network addresses, Network News Transfer Protocol (**NNTP**) is an application protocol used for transporting Usenet news articles (netnews) between news servers and for reading and posting articles by end user client applications, the protocol for moving USENET news articles around, and HTTP, the protocol for fetching pages on the World Wide Web, and many others.

The TCP/IP model with some protocols we will study.



- **Comparison of the OSI and TCP/IP Reference Models**

Similarities

- Both are based on the concept of a stack of independent protocols.
- Both of them have transport layer to provide an end-to-end, network-independent transport service

Differences

- In OSI, a layer's service definition tells what the layer does, and a layer's interface tells the processes above it how to access it, without explaining how the layers works inside.
- The TCP/IP model did not clearly distinguish between service, interface of the layers.
- As a consequence, the protocols in the OSI model are better hidden than in the TCP/IP model and can be replaced relatively easily as the technology changes.
- OSI model strives to describe the general model of network protocols, while TCP/IP model is specific as it is defined after the real implementations. So TCP/IP model is not useful to describe other protocols.

- The hybrid reference model

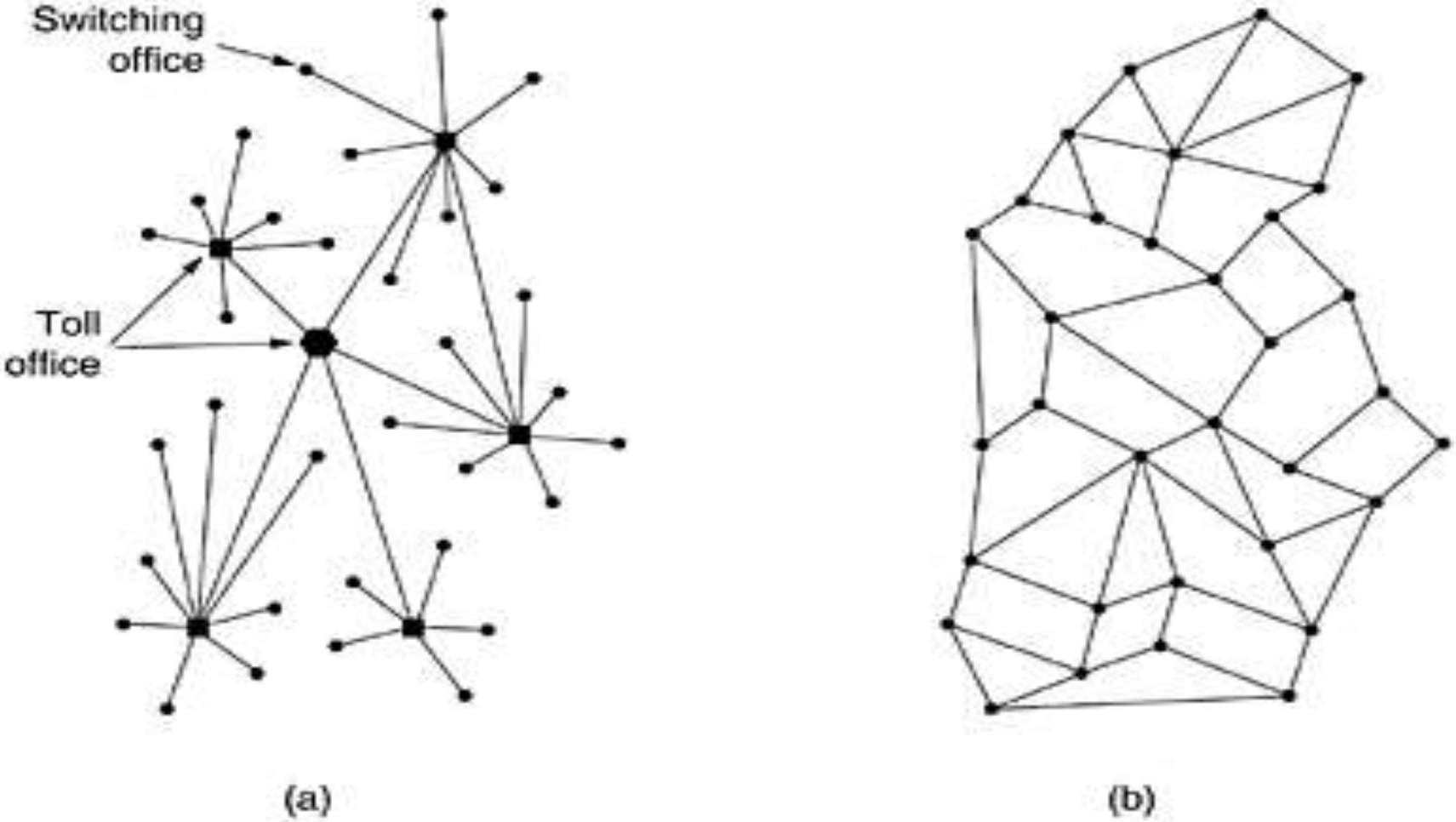
5	Application layer
4	Transport layer
3	Network layer
2	Data link layer
1	Physical layer

Three concepts are central to the OSI model:

1. Services
2. Interfaces
3. Protocols

- The **service** definition tells what the layer does, not how entities above it access it or how the layer works. It defines the layer's semantics.
- A layer's **interface** tells the processes above it how to access it. It specifies what the parameters are and what results to expect. It, too, says nothing about how the layer works inside.
- Finally, the peer **protocols** used in a layer are the layer's own business. It can use any protocols it wants to, as long as it gets the job done (i.e., provides the offered services). It can also change them without affecting software in higher layers.

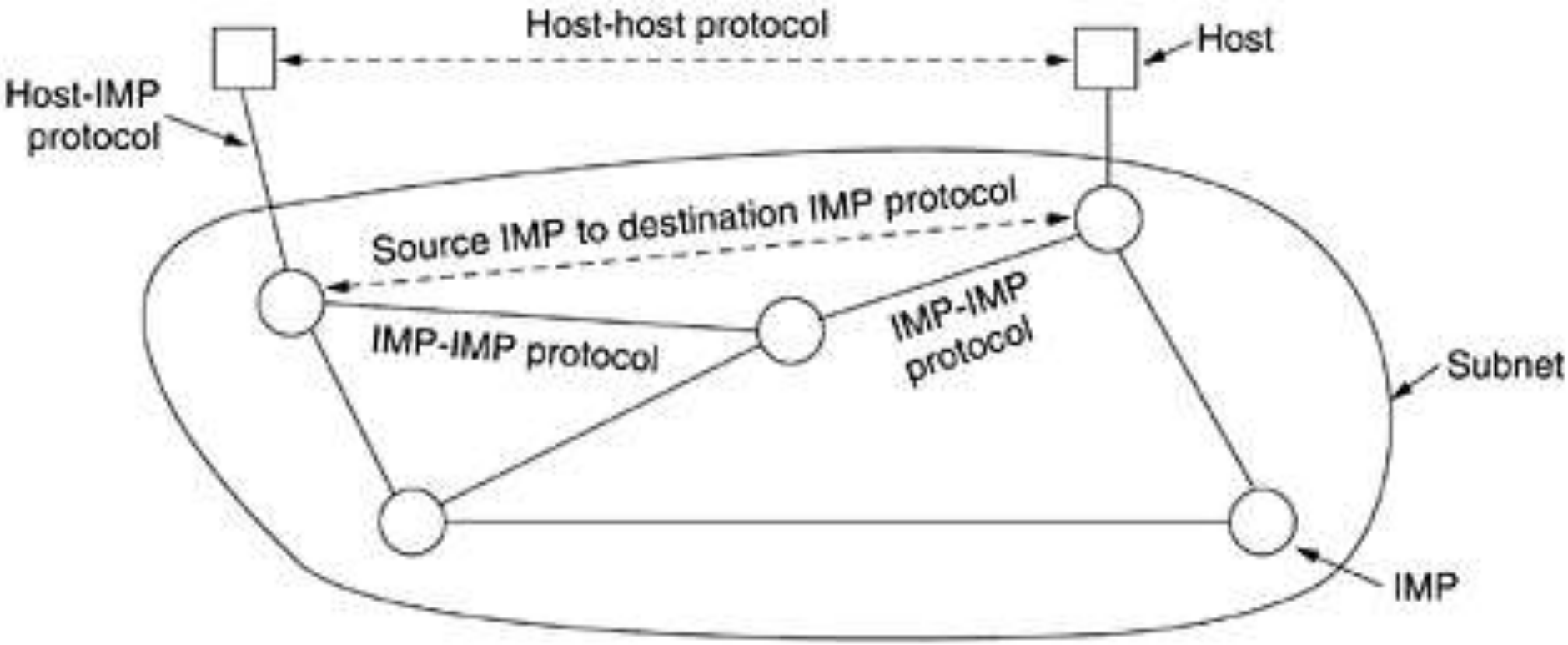
IV. Example Networks — The Internet



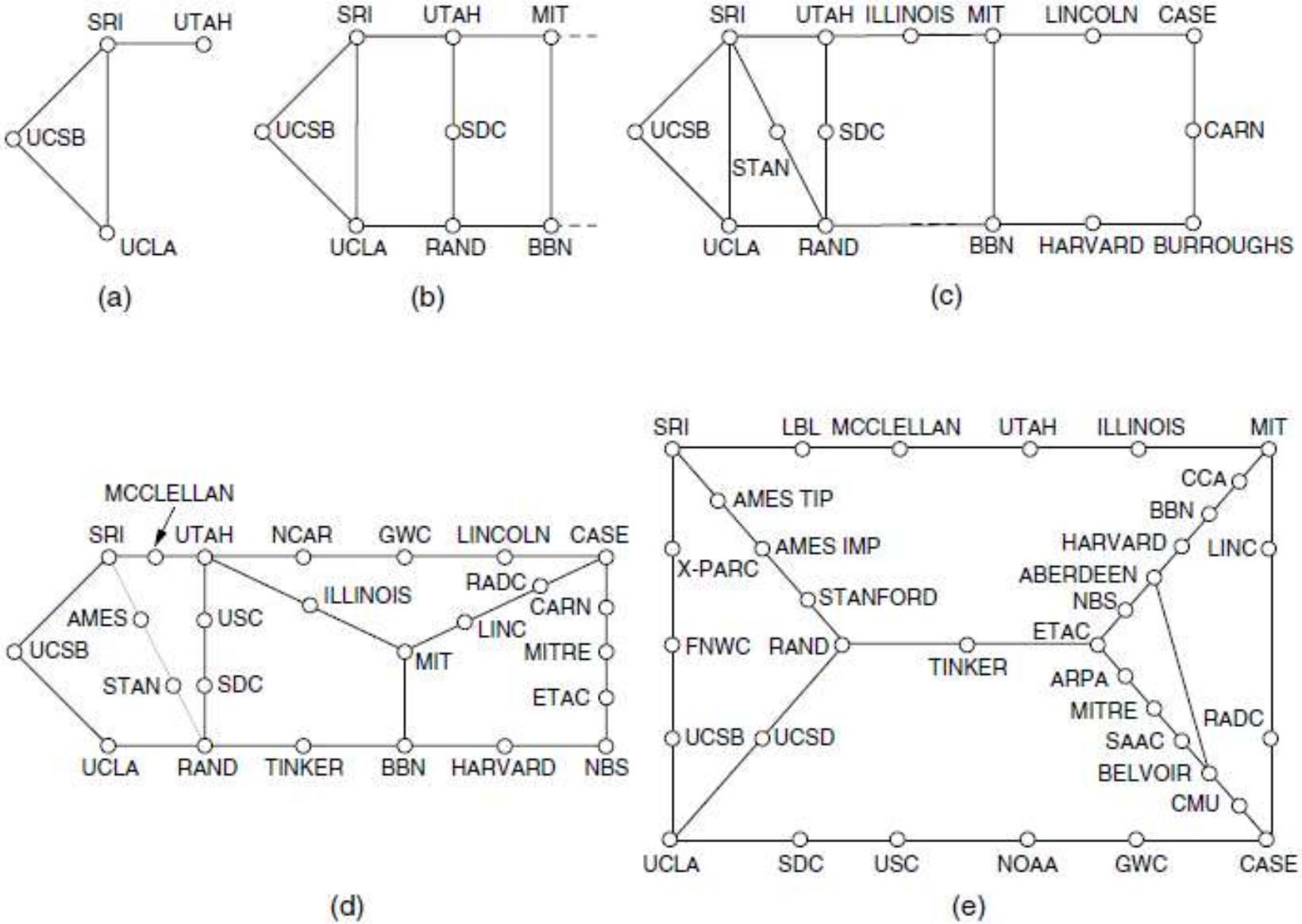
(a) Structure of the telephone system.

(b) Baran's proposed distributed switching system.

• Example Networks — The Internet 2

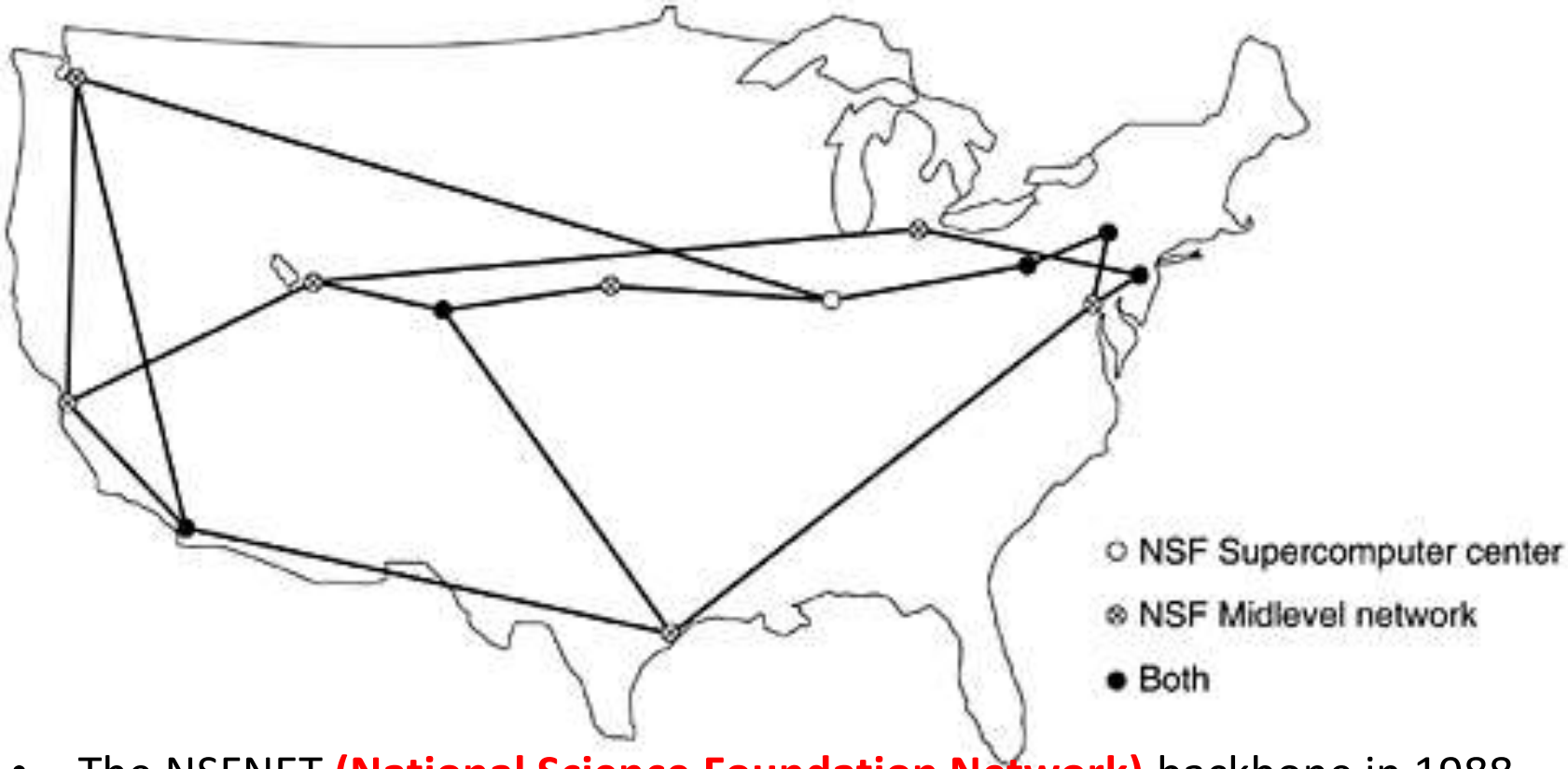


The original ARPANET design (IMP - Interface Message Processors- minicomputers)



Growth of the ARPANET. (a) December 1969. (b) July 1970. (c) March 1971. (d) April 1972. (e) September 1972.

• Example Networks — The Internet 3

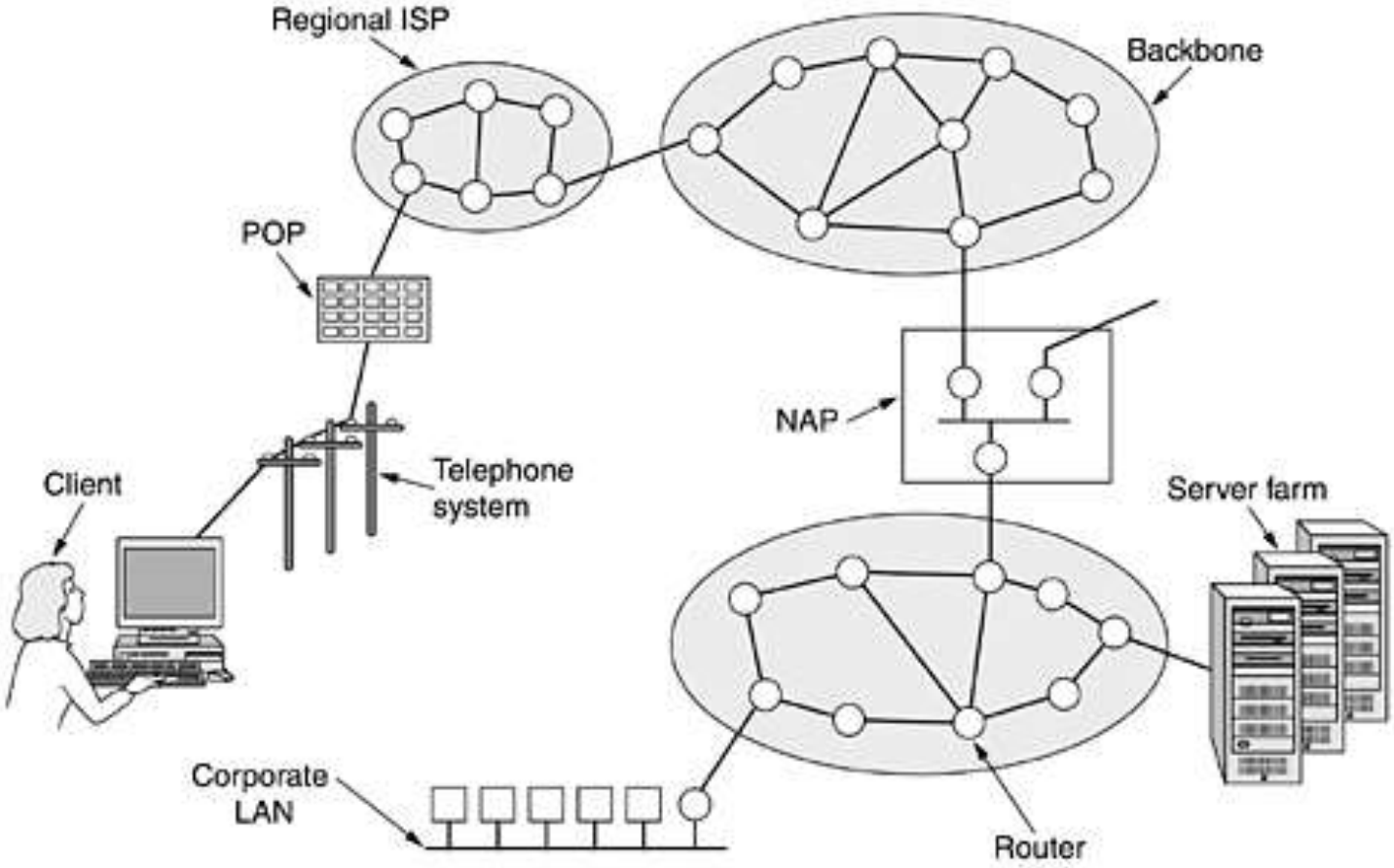


- The NSFNET **(National Science Foundation Network)** backbone in 1988

Microcomputer called a **fuzzball.**

- **The fuzzballs** were connected with 56-kbps leased lines and formed the subnet, the same hardware technology the ARPANET used.
- The software technology was different however: the fuzzballs spoke TCP/IP right from the start, making it the first TCP/IP WAN.

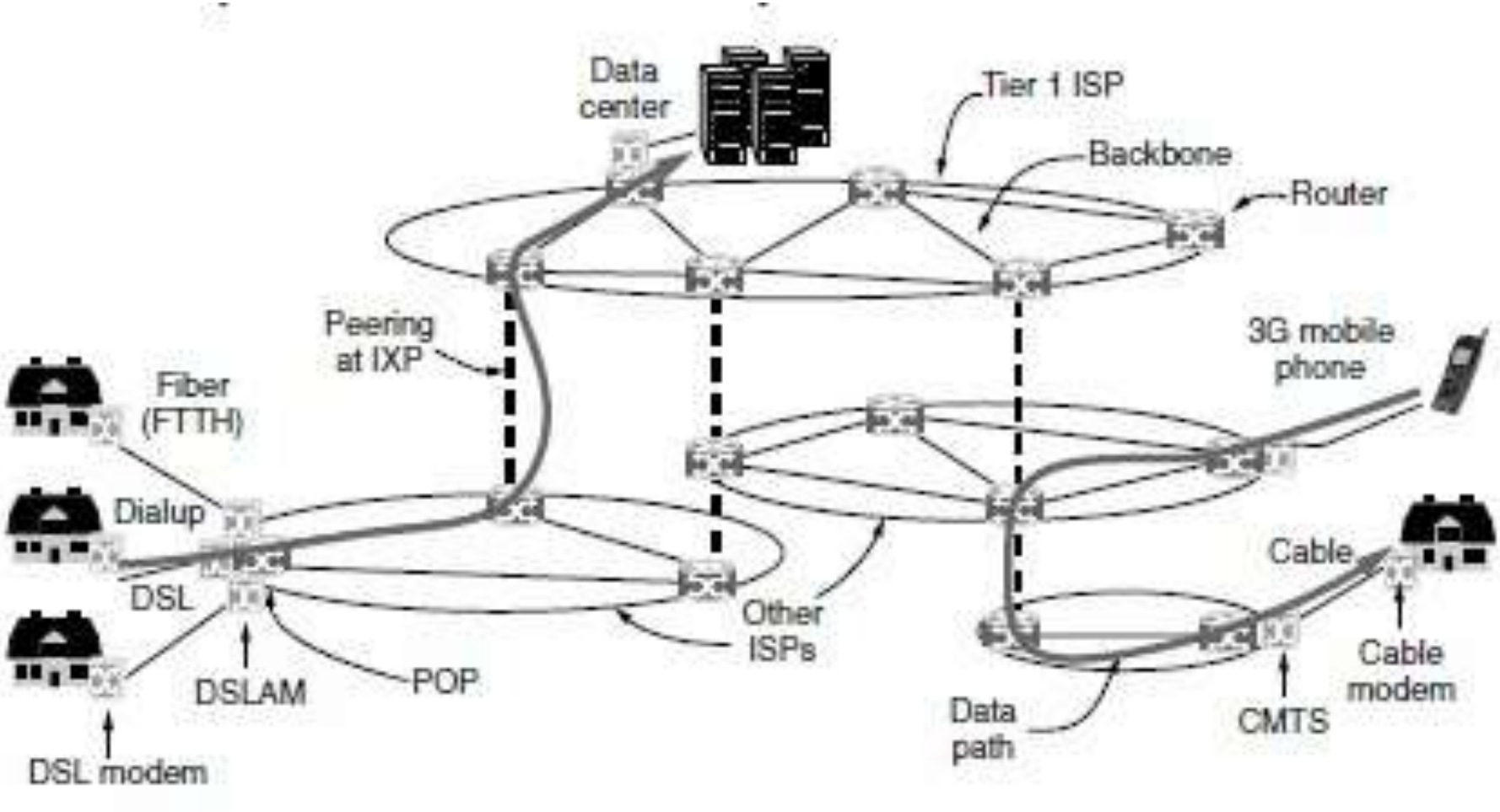
- Example Networks —— The Internet 4



Architecture of the Internet

POP: Point Of Presence
ISP: Internet Service Provider
NAP: Network Access Point

Overview of the Internet architecture
internet exchange points- ixps
fiber to home -ftth
DSLAM (Digital Subscriber Line Access Multiplexer)
CMTS: Cable Modem Termination System



- A common way to connect to an ISP is to use the phone line to your house, in which case your phone company is your ISP.
- **DSL- Digital Subscriber Line,** reuses the telephone line that connects to your house for digital data transmission.
- The computer is connected to a device called a **DSL modem that** converts between digital packets and analog signals that can pass unrestricted over the telephone line.
- At the other end, a device called a **DSLAM (Digital Subscriber Line Access Multiplexer) converts between signals and packets.**
- Several other popular ways to connect to an ISP are – **Dial-up-modem** for ‘‘modulator demodulator’’ and refers to any device that converts between digital bits and analog signals.

- Another method is to send signals over the cable TV system. Like DSL, this is a way to reuse existing infrastructure.
- Cable Modem and the device at the cable head end is called the CMTS (Cable Modem Termination System).
- Internet access at much greater than dial-up speeds is called broadband.
- ISP networks may be regional, national, or international in scope.
- ISPs connect their networks to exchange traffic at **IXPs (Internet eXchange Points).**
- Internet used to be that a machine was on the Internet if it:

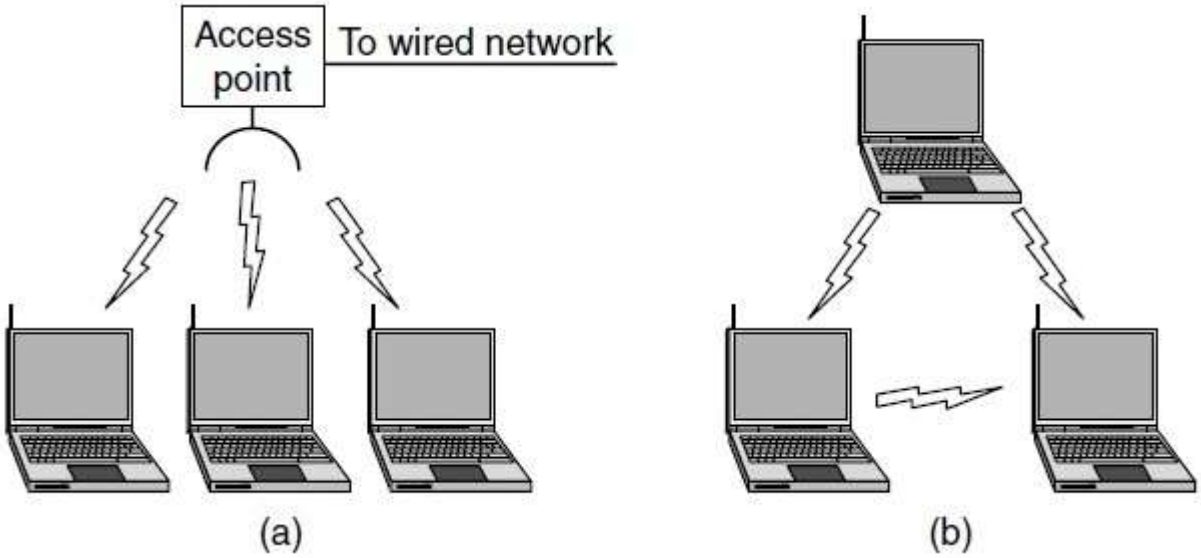
  (1) ran the TCP/IP protocol stack;

  (2) had an IP address; and

  (3) could send IP packets to all the other machines on the Internet.

Wireless LANs: 802.11

- **Wi-Fi -** Wireless Fidelity
- 802.11 networks are made up of clients, such as laptops and mobile phones, and infrastructure called **APs (access points) that is installed in buildings.** Access points are sometimes called **base stations.**
- The **access points** connect to the wired network, and all communication between clients goes through an access point.
- Two Computers in an office without an access point. This arrangement is called an **ad hoc network.**

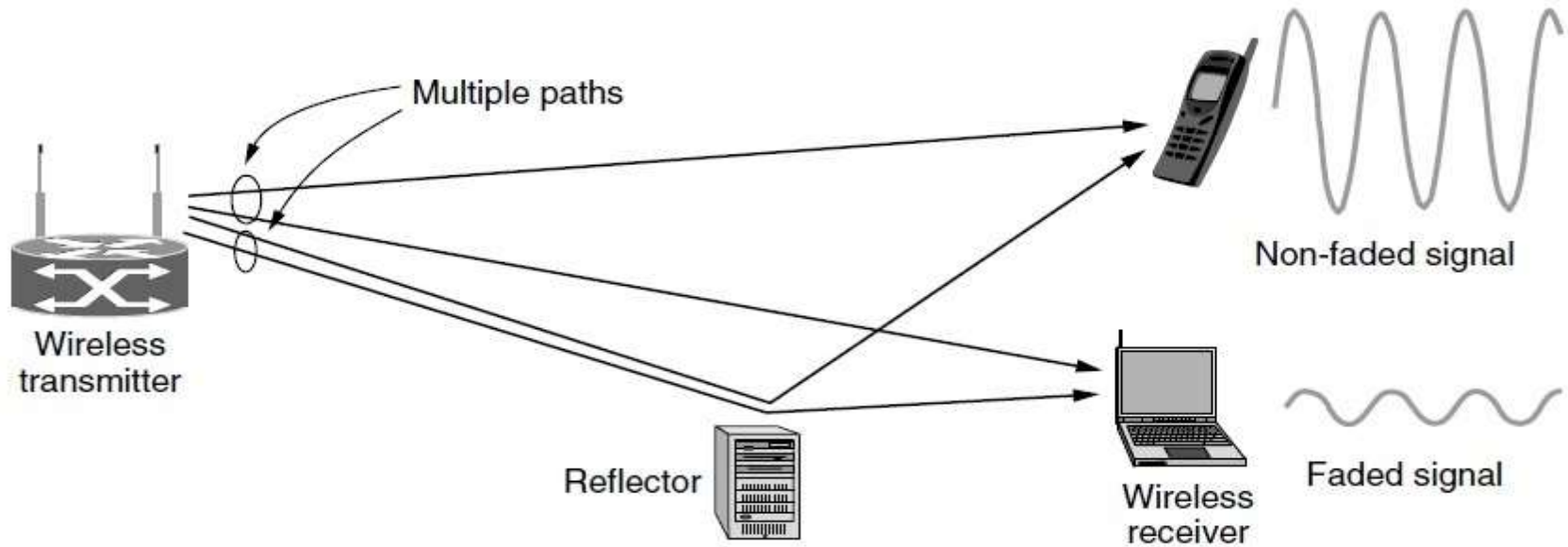


(a) Wireless network with an access point. (b) Ad hoc network

Multipath Fading :

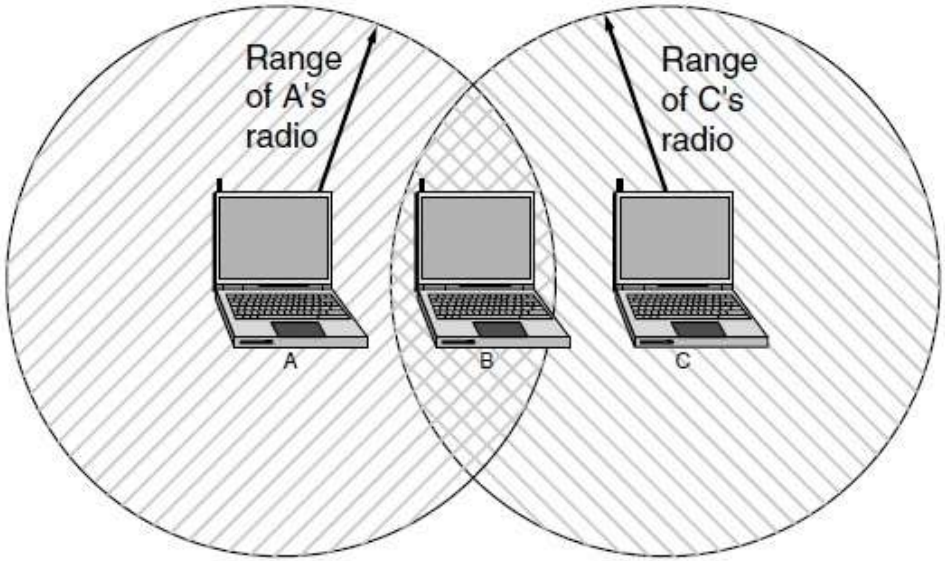
- Frequencies used for 802.11, radio signals can be reflected off solid objects so that multiple echoes of a transmission may reach a receiver along different paths.
- The echoes can cancel or reinforce each other, causing the received signal to fluctuate greatly-**multipath fading.**
- The key idea for overcoming variable wireless conditions or **multipath fading** is **path diversity**, or the sending of information **along multiple, independent paths**.
- The information is likely to be received even if one of the paths happens to be poor due to a fade. These independent paths are typically built into the digital modulation scheme at the physical layer. Options include using different frequencies across the allowed band, following different spatial paths between different pairs of antennas, or repeating bits over different periods of time.

Multipath fading

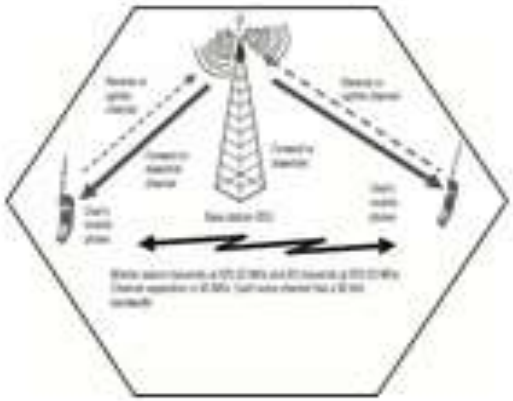


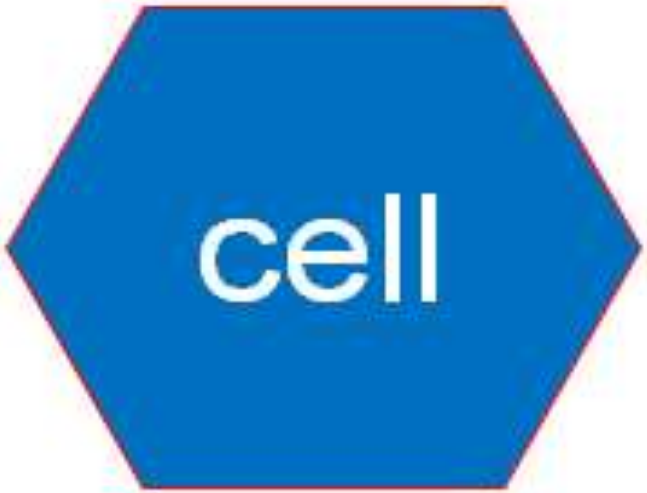
- The 802.11a (1999) and 802.11g (2003-WLAN) standards switched to a different modulation scheme called **OFDM (Orthogonal Frequency Division Multiplexing).**
- **It divides a wide band** of spectrum into many narrow slices over which different bits are sent in parallel.
- The latest version is 802.11n (2009). It uses wider frequency bands and up to four antennas per computer to achieve rates up to 450 Mbps.

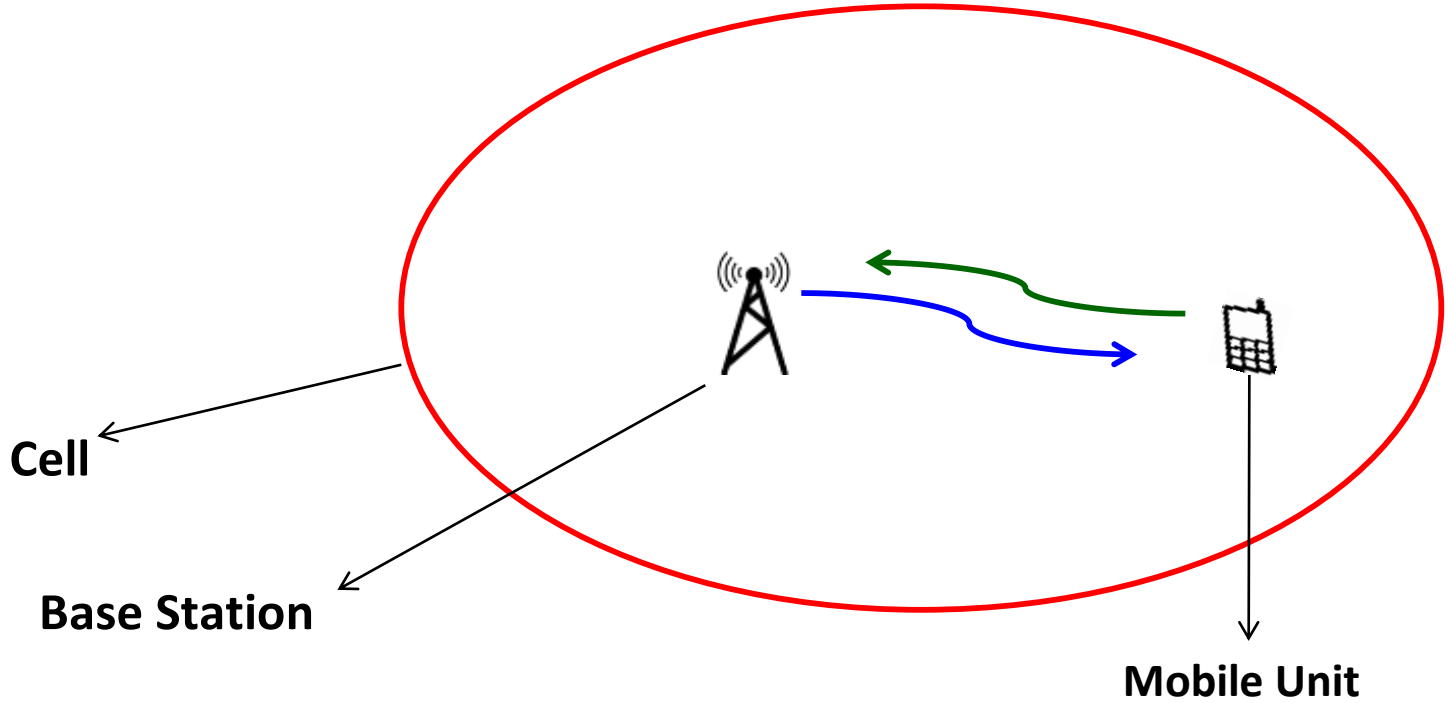
- 802.11 radios also have to deal with the **problem** of **multiple transmissions** that are **sent at the same time will collide**, which may interfere with reception. To handle this problem, 802.11 uses a **CSMA (Carrier Sense Multiple Access).**
- To handle this problem, 802.11 uses a CSMA scheme and ALOHA.

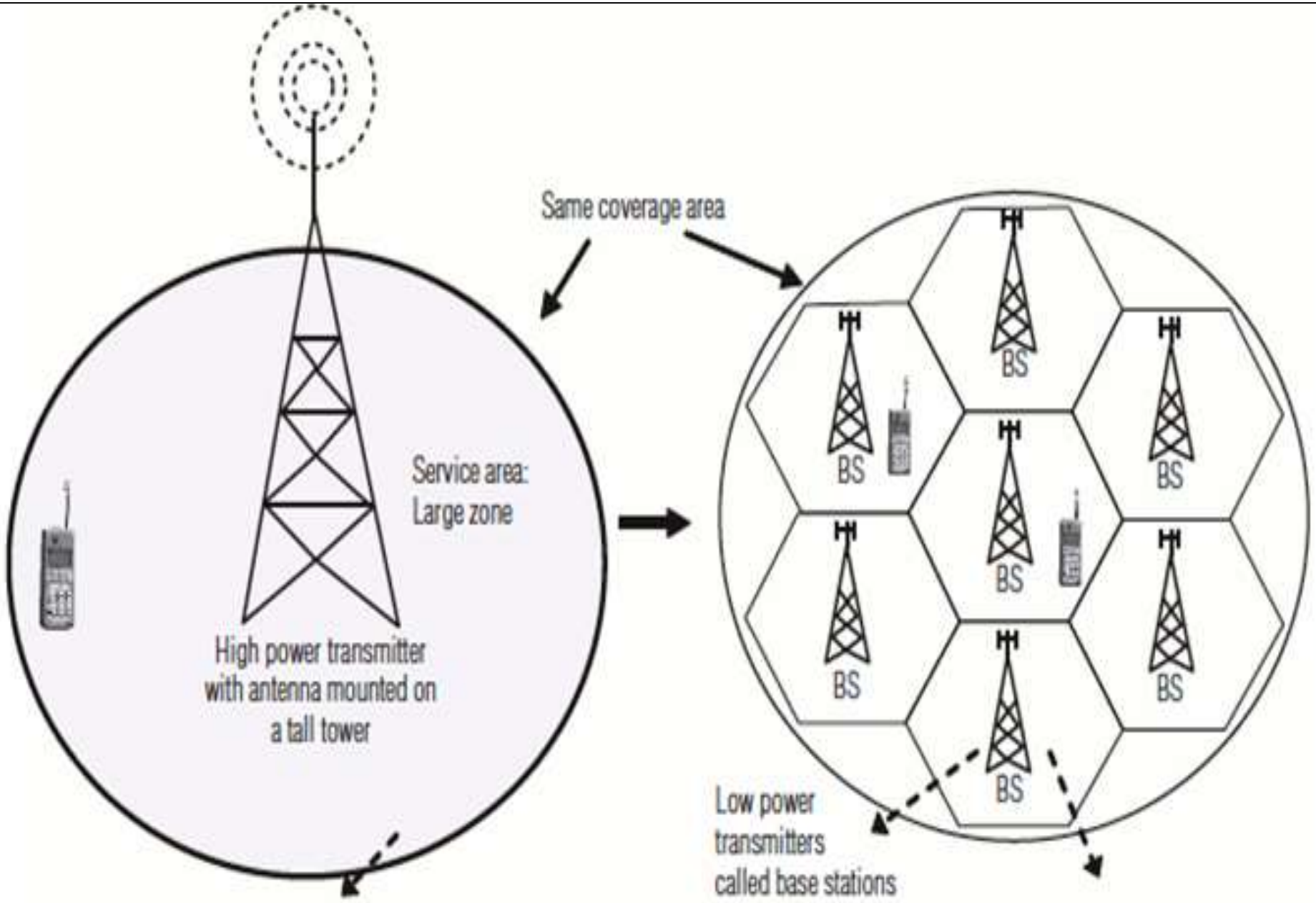


The range of a single radio may not cover the entire system



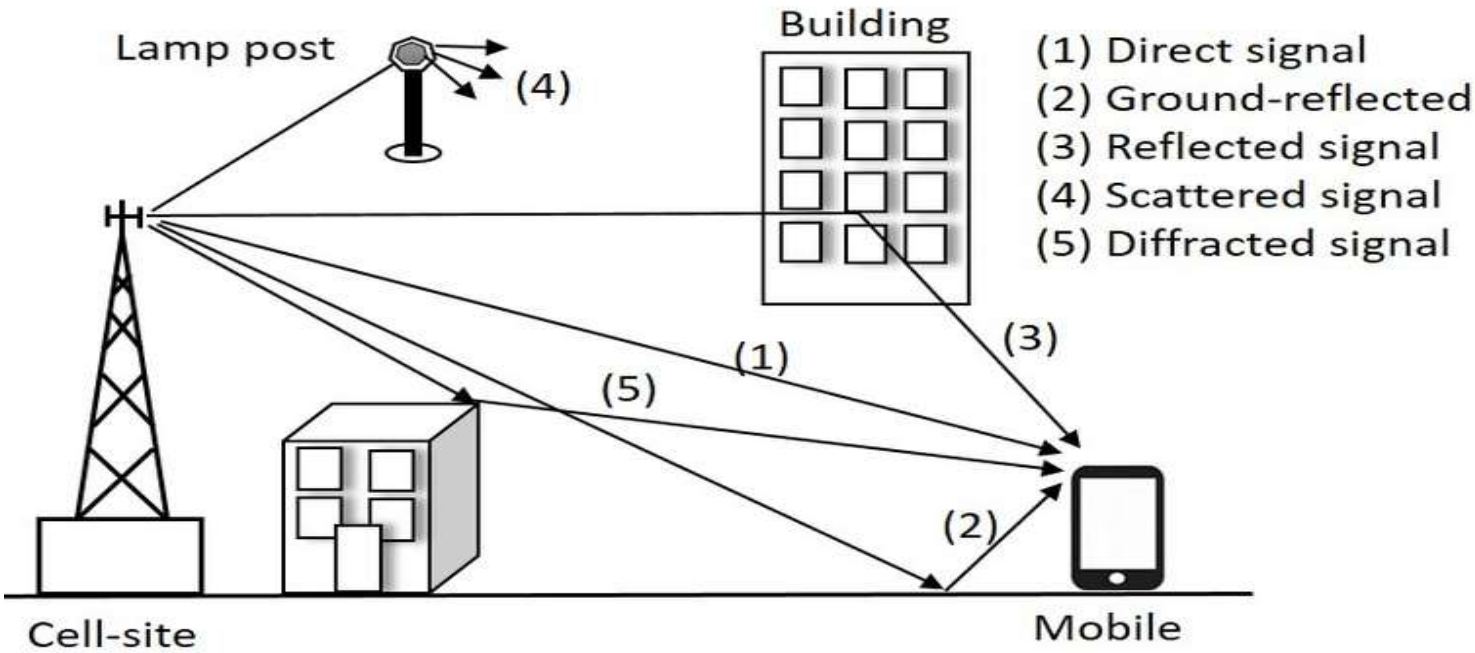

cell

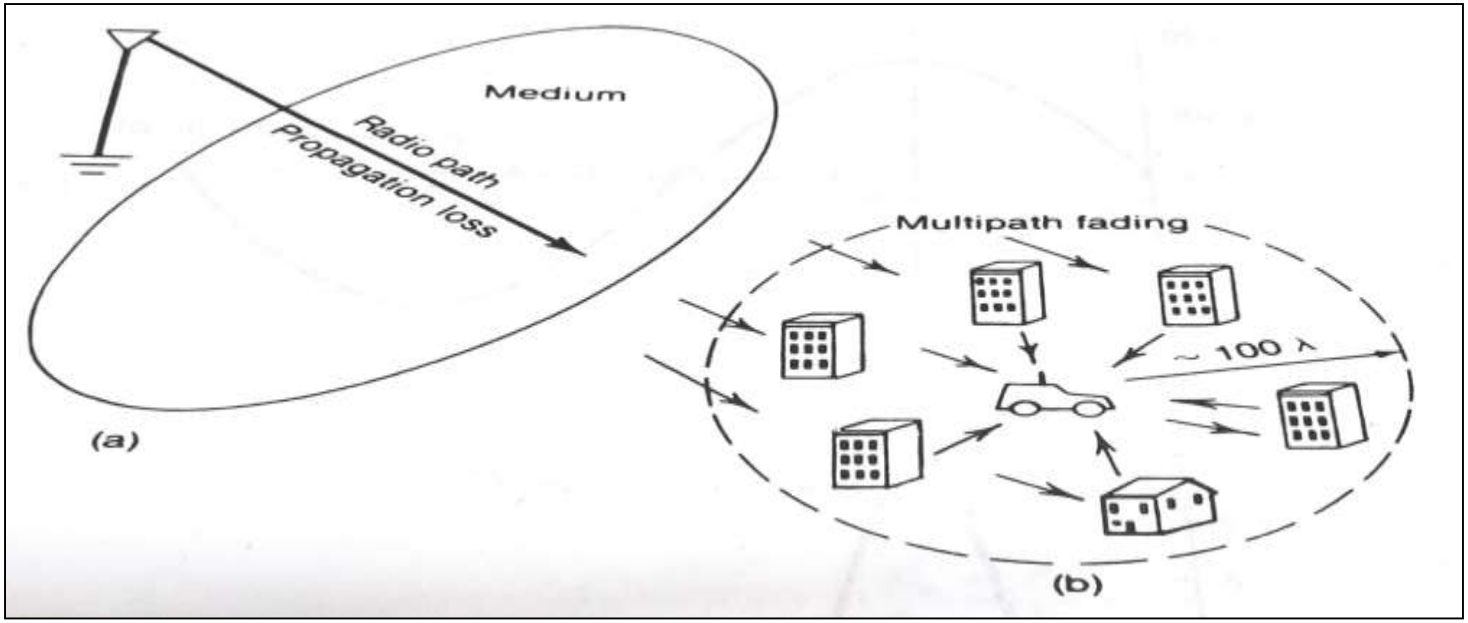

Cell
Base Station
Mobile Unit



(a) Early mobile radio system: Large zone

(b) Cellular system: small zone


Lamp post
(4)
Building
(1) Direct signal
(2) Ground-reflected
(3) Reflected signal
(4) Scattered signal
(5) Diffracted signal
(1)
(3)
(5)
(2)
Cell-site
Mobile


Medium
Radio path
Propagation loss
(a)
Multipath fading
~ 100 λ
(b)

- Another **problem** is that of **mobility**. If a mobile client is moved away from the access point it is using and into the range of a different access point, some way of **handing it off** is needed.
- The solution is that an 802.11 network can consist of **multiple cells**, each with its **own access point**, and a **distribution system** that connects the cells.
- The **distribution system** is often switched Ethernet, but it can use any technology.
- **problem of security:** Since wireless transmissions are broadcast, it is easy for nearby computers to receive packets of information that were not intended for them.
- To prevent this, the 802.11 standard included an **Encryption Scheme** known as **WEP (Wired Equivalent Privacy).**
- The idea was to make wireless security like that of wired security.
- **WEP** has since been replaced with newer schemes that have different cryptographic details in the **802.11i** standard, also called **WPA- WiFi Protected Access,** but now replaced by **WPA2.**

- 802.11 has caused a revolution in wireless networking that is set to continue.
- Beyond buildings, it is starting to be installed in trains, planes, boats, and automobiles so that people can surf the Internet wherever they go.
- Mobile phones and all manner of consumer electronics, from game consoles to digital cameras, can communicate with it.

- # Network Standardization

Number	Topic
802.1	Overview and architecture of LANs
802.2 ↓	Logical link control
802.3 *	Ethernet
802.4 ↓	Token bus (was briefly used in manufacturing plants)
802.5	Token ring (IBM's entry into the LAN world)
802.6 ↓	Dual queue dual bus (early metropolitan area network)
802.7 ↓	Technical advisory group on broadband technologies
802.8 †	Technical advisory group on fiber optic technologies
802.9 ↓	Isochronous LANs (for real-time applications)
802.10 ↓	Virtual LANs and security
802.11 *	Wireless LANs
802.12 ↓	Demand priority (Hewlett-Packard's AnyLAN)
802.13	Unlucky number. Nobody wanted it
802.14 ↓	Cable modems (defunct: an industry consortium got there first)
802.15 *	Personal area networks (Bluetooth)
802.16 *	Broadband wireless
802.17	Resilient packet ring

The IEEE 802 series standards. The important ones are marked with *. The ones marked with ↓ are hibernating. The one marked with † gave up and disbanded itself.

PHYSICAL LAYER:

GUIDED TRANSMISSION MEDIA

1. Magnetic Media
2. Twisted Pairs
3. Coaxial Cable
4. Power Lines
5. Fiber Optics

Magnetic Media

- One of the most common ways to transport data from one computer to another is to write them onto magnetic tape or removable media (e.g., recordable DVDs), physically transport the tape or disks to the destination machine, and read them back in again.
- An industry-standard **Ultrium tape** can hold 800 gigabytes.
- A box 60 × 60 × 60 cm can hold about 1000 of these tapes, for a total capacity of 800 terabytes, or 6400 terabits (6.4 petabits).
- The cost of an **Ultrium** tape is around $40 when bought in bulk. A tape can be reused at least 10 times, so the tape cost is maybe $4000 per box per usage.
- *Never underestimate the bandwidth of a station full of tapes hurtling down the highway.*

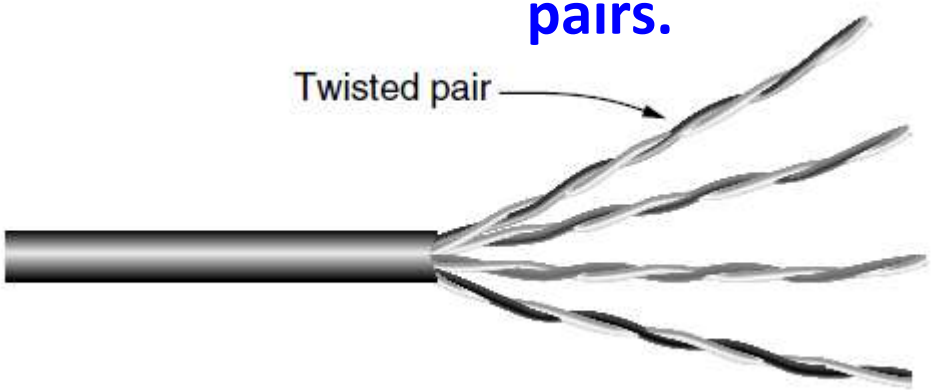
- Although the bandwidth characteristics of magnetic tape are excellent, the delay characteristics are poor.
- Transmission time is measured in minutes or hours, not milliseconds.
- For many applications an online connection is needed.

2. Twisted Pairs

- One of the oldest and still most common transmission media is **twisted pair**.
- A twisted pair consists of two insulated copper wires, typically about 1 mm thick.
- The wires are twisted together in a helical form, just like a DNA molecule.
- Twisting is done because two parallel wires constitute a fine antenna.
- When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively.

- The most common application of the twisted pair is the telephone system.
- Twisted pairs can run several kilometers without amplification, but for longer distances the signal becomes too attenuated and repeaters are needed.
- Twisted pairs can be used for transmitting either analog or digital information.
- The bandwidth depends on the thickness of the wire and the distance traveled, but several megabits/sec can be achieved for a few kilometers in many cases.
- Due to their adequate performance and low cost, twisted pairs are widely used.
- A **category 5 twisted pair** consists of two insulated wires gently twisted together. Four such pairs are typically grouped in a plastic sheath to protect the wires and keep them together.

Category 5 UTP (Unshielded Twisted pair) cable with four twisted pairs.



- **100-Mbps Ethernet** uses two (out of the four) pairs, one pair for each direction.
- To reach higher speeds, 1-Gbps Ethernet uses all four pairs in both directions (**full- duplex links**- used in both directions at the same time, like a two-lane road) simultaneously.
- **Half-duplex links** - used in either direction, but only one way at a time, like a single-track railroad line.
- A third category consists of links that allow traffic in only one direction, like a one-way street. They are called **simplex links**.

- **Category 3 cables** that uses the same connector, but has more twists per meter. More twists result in less crosstalk and a better-quality signal over longer distances, making the cables more suitable for high-speed computer communication, especially 100-Mbps and 1-Gbps Ethernet LANs.
- **Category 6 or even Category 7-** **These** categories has more stringent specifications to handle signals with greater bandwidths.
- Some cables in Category 6 and above are rated for signals of 500 MHz and can support the 10-Gbps links that will soon be deployed.
- Through **Category 6,** these wiring types are referred to as **UTP (Unshielded Twisted Pair) as they consist simply of wires and insulators.**
- **In contrast to these, Category 7 cables** have shielding on the individual twisted pairs, as well as around the entire cable (but inside the plastic protective sheath). Shielding reduces the susceptibility to external interference and crosstalk with other nearby cables to meet demanding performance specifications.
- **Expensive shielded twisted pair cables** that **IBM** introduced in the early 1980s, but which did not prove popular outside of IBM installations.

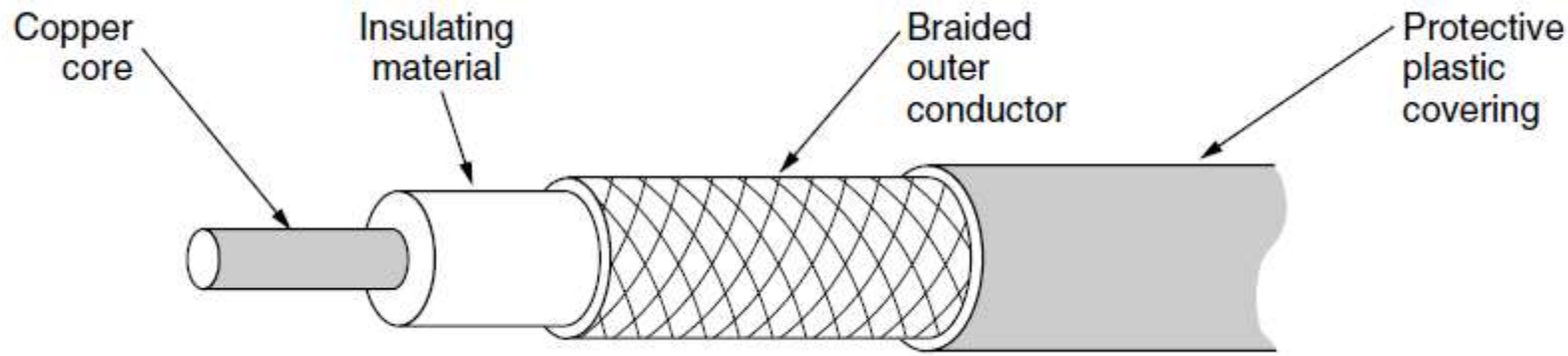
3. Coaxial Cable

- It has better shielding and greater bandwidth than unshielded twisted pairs, so it can span **longer distances at higher speeds**.

Two kinds of coaxial cable are widely used.

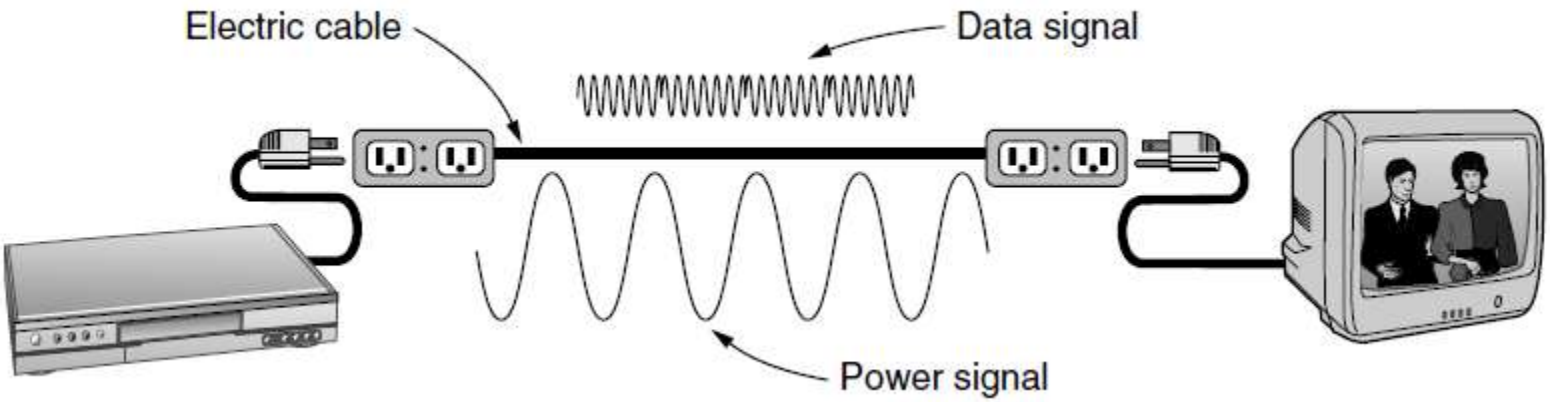
- One kind, **50-ohm cable**, is commonly used when it is intended for digital transmission from the start.
- The other kind, **75-ohm cable**, is commonly used for analog transmission and cable television.
- Starting in the mid- 1990s, cable TV operators began to provide Internet access over cable, which has made **75-ohm cable** more important for data communication.
- A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh.
- The outer conductor is covered in a protective plastic sheath(cover).

A cutaway view of a coaxial cable



- The construction and shielding of the coaxial cable give it a good combination of **high bandwidth** and **excellent noise immunity**.
- The **bandwidth** possible depends on the cable quality and length. Modern cables have a bandwidth of up to a few GHz.
- **Coaxial cables** used to be widely used within the telephone system for long-distance lines but have now largely been replaced by fiber optics on long haul routes.
- **Coax** is still widely used for cable television and metropolitan area networks.

4. Power Lines



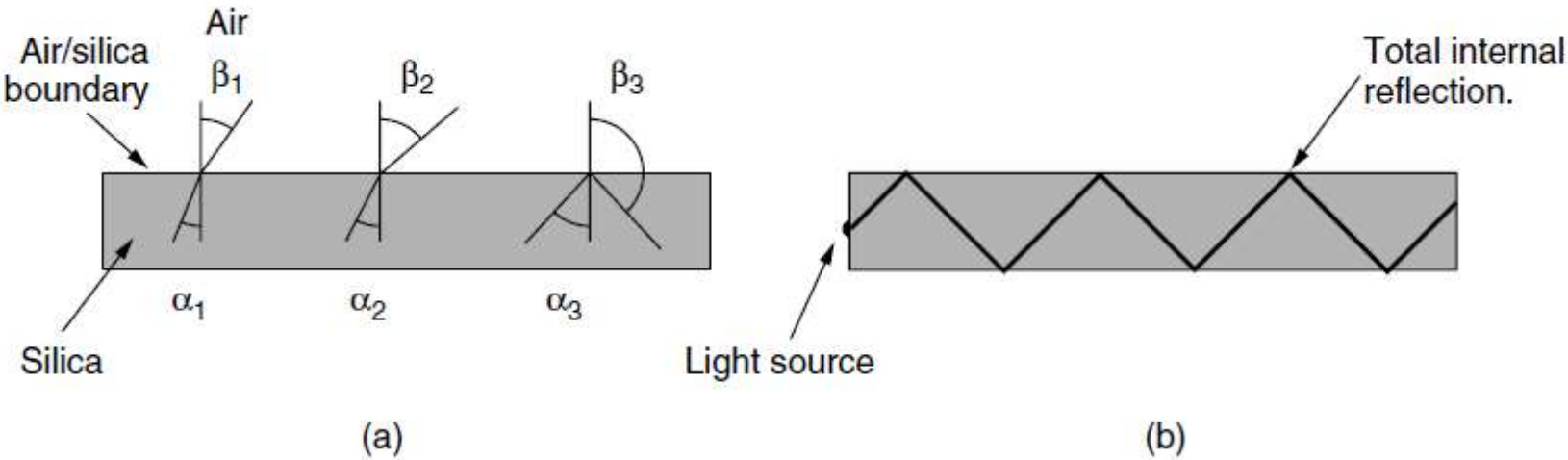
A network that uses household electrical wiring

5. Fiber Optics

An optical transmission system has three key components:

1. The Light Source: A pulse of light indicates a 1 bit and the absence of light indicates a 0 bit.
2. The Transmission Medium is an ultra-thin fiber of glass
3. The detector generates an electrical pulse when light falls on it

- By attaching a light source to one end of an optical fiber and a detector to the other, we have a unidirectional data transmission system that accepts an electrical signal, converts and transmits it by light pulses, and then reconverts the output to an electrical signal at the receiving end.
- This transmission system would leak light and be useless in practice.



(a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles.

(b) Light trapped by total internal reflection and can propagate for many kilometers with virtually no loss.

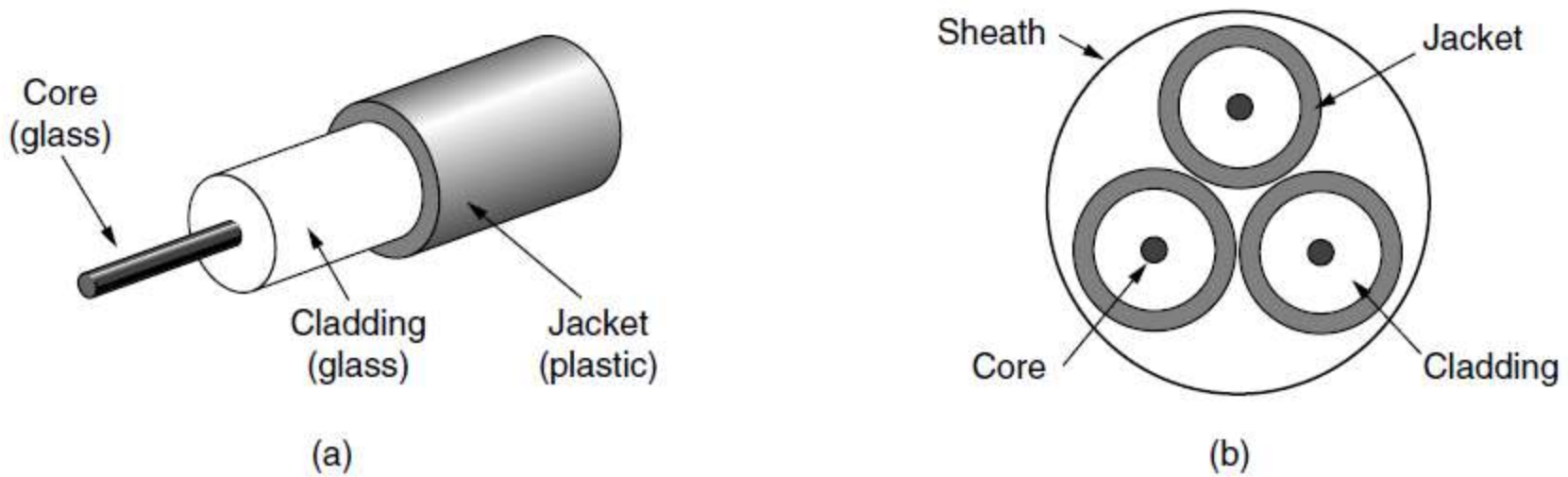
- Single-mode fibers can transmit data at 100 Gbps for 100 km without amplification.
- Even higher data rates have been achieved in the laboratory for shorter distances.

Transmission of Light Through Fiber

- Optical fibers are made of glass, which, in turn, is made from sand, an inexpensive raw material available in unlimited amounts.
- Glassmaking was known to the ancient Egyptians, but their glass had to be no more than 1 mm thick or the light could not shine through.
- Glass transparent enough to be useful for windows was developed during the Renaissance (regeneration).
- The glass used for modern optical fibers is so transparent that if the oceans were full of it instead of water, the seabed would be as visible from the surface as the ground is from an airplane on a clear day.

Fiber Cables

- Fiber optic cables are similar to coax, except without the braid.
- At the center is the glass core through which the light propagates. In **multimode fibers**, the core is typically 50 microns in diameter, about the thickness of a human hair.
- In **single-mode fibers**, the core is 8 to 10 microns.



(a) Side view of a single fiber.
(b) End view of a sheath with three fibers

Fibers can be connected in three different ways.

- First, they can terminate in connectors and be plugged into fiber sockets.
- Connectors lose about 10 to 20% of the light, but they make it easy to reconfigure systems.
- Second, they can be spliced mechanically.
- Mechanical splices just lay the two carefully cut ends next to each other in a special sleeve and clamp them in place.
- Alignment can be improved by passing light through the junction and then making small adjustments to maximize the signal.
- Mechanical splices take trained personnel about 5 minutes and result in a 10% light loss.
- Third, two pieces of fiber can be fused (melted) to form a solid connection. A fusion splice is almost as good as a single drawn fiber, but even here, a small amount of attenuation occurs.

- Two kinds of light sources are typically used to do the signaling. These are LEDs (Light Emitting Diodes) and semiconductor lasers.

Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multi-mode	Multi-mode or single-mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

A comparison of semiconductor diodes and LEDs as light sources.

Comparison of Fiber Optics and Copper Wire

- Fiber can handle much higher bandwidths than copper.
- Due to the low attenuation, repeaters are needed only about every 50 km on long lines, versus about every 5 km for copper, resulting in a big cost saving.
- Fiber also has the advantage of not being affected by power surges, electromagnetic interference, or power failures. Nor is it affected by corrosive chemicals in the air, important for harsh factory environments.
- Fiber has less weight compare to copper.

WIRELESS TRANSMISSION

- **The Electromagnetic Spectrum**
- **Radio Transmission**
- **Microwave Transmission**
- **Infrared Transmission**
- **Light Transmission**

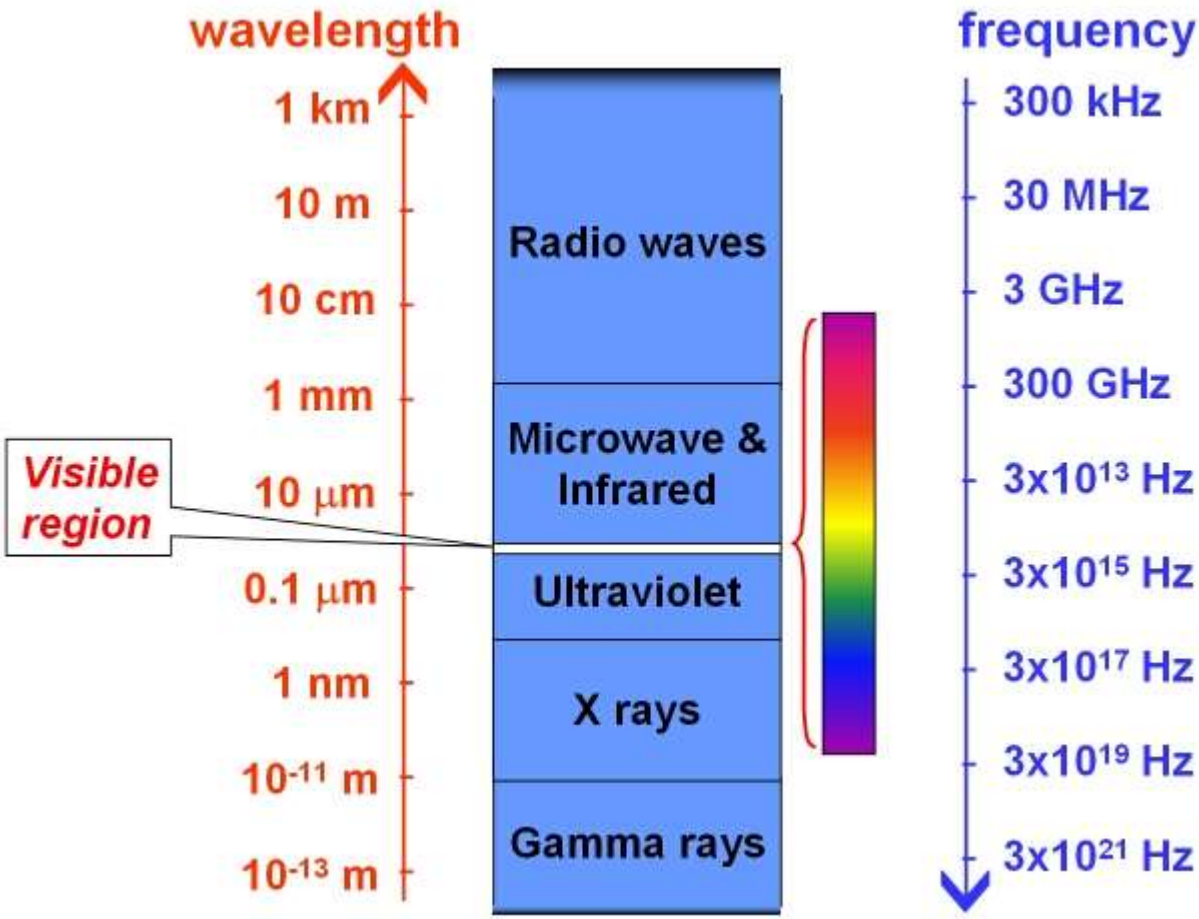
People who need to be online for all time. For these mobile users Twisted Pairs, Coaxial Cable and Fiber Optics are of no use.

For mobile users , Wireless communication is the solution.

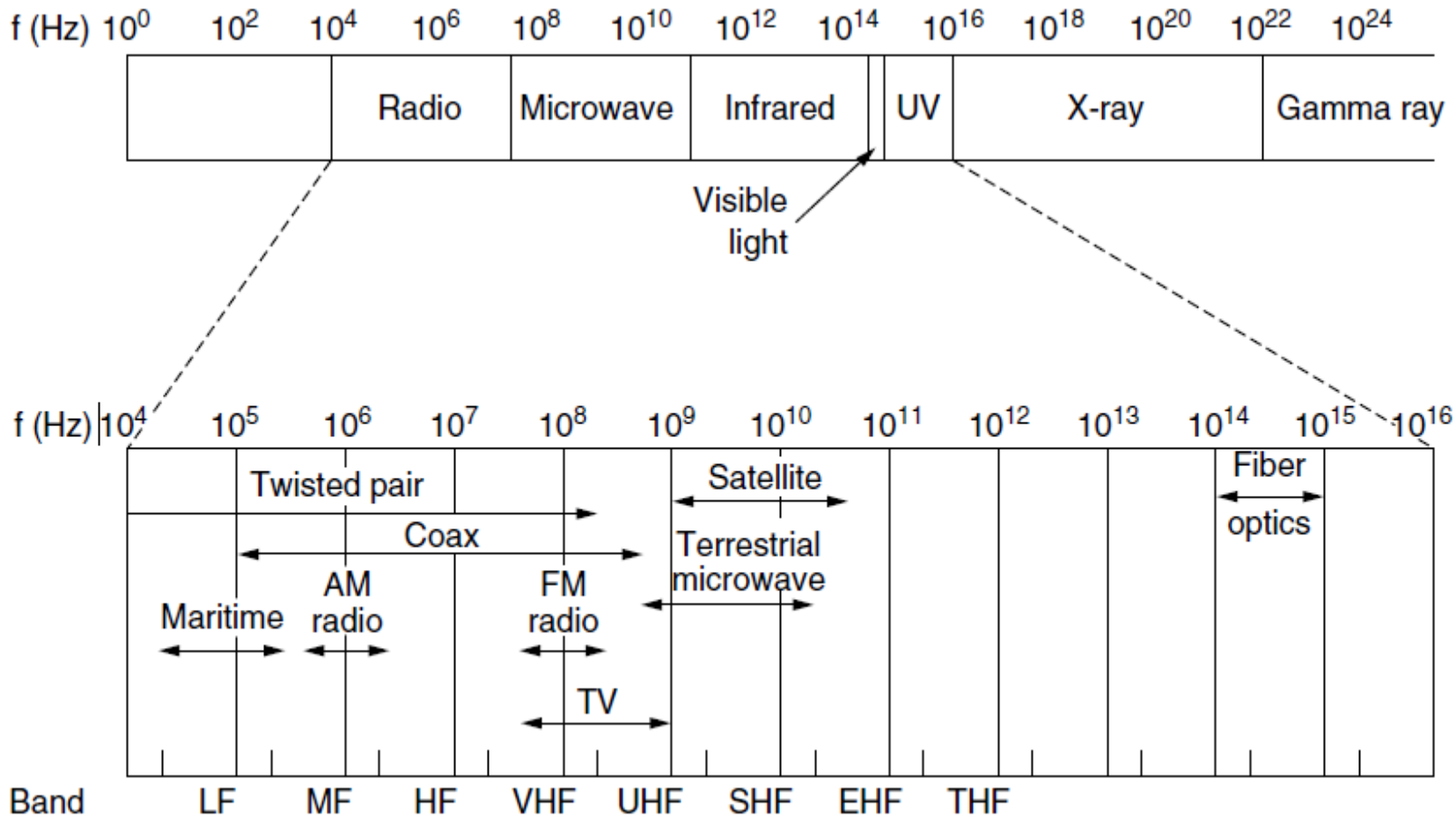
The Electromagnetic Spectrum

- The number of oscillations per second of a wave is called its **frequency, *f, and is measured in Hz (in honor of Heinrich*** Hertz).
- The distance between two consecutive maxima (or minima) is called the **wavelength,** which is universally designated by the Greek letter λ (lambda).
- First form of spread spectrum, **frequency hopping spread spectrum**, the transmitter hops from frequency to frequency hundreds of times per second. It is popular for military communication because it makes transmissions hard to detect and next to impossible to jam. It also offers good resistance to multipath fading and narrowband interference because the receiver will not be stuck on an impaired frequency for long enough to shut down communication.

ELECTROMAGNETIC SPECTRUM



The electromagnetic spectrum and its uses for communication.



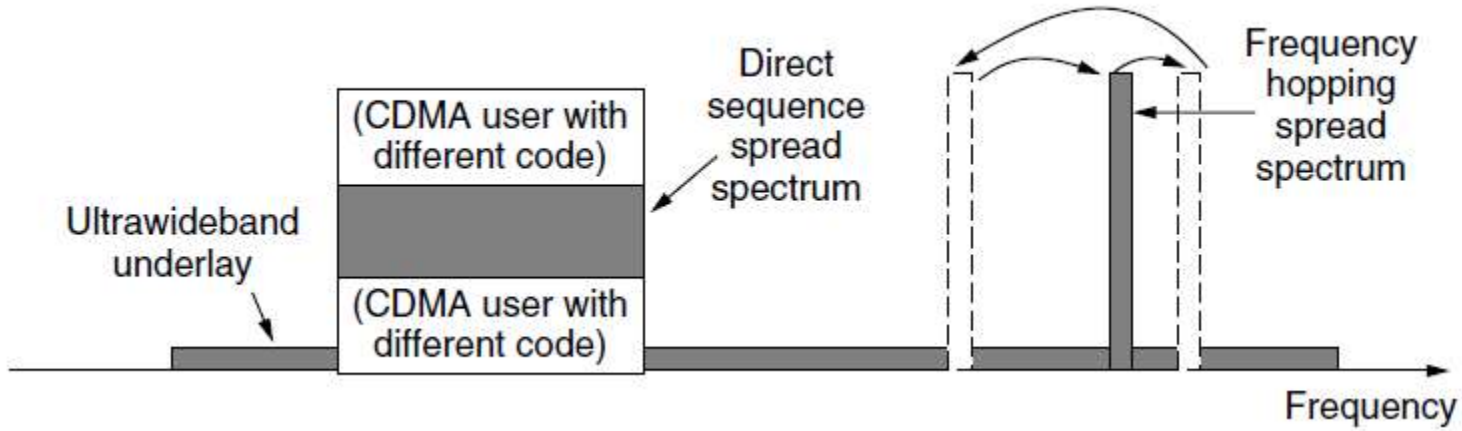
Very high frequency	VHF
Ultra high frequency	UHF
Super high frequency	SHF
Extremely high frequency	EHF
Terahertz or Tremendously high frequency	THz or THF

A second form of spread spectrum, **direct sequence spread spectrum,** uses a code sequence to spread the data signal over a wider frequency band.

- It is widely used commercially as a spectrally efficient way to let multiple signals share the same frequency band.
- These signals can be given different codes, a method called **CDMA (Code Division Multiple Access).**

A third method of communication with a wider band is **UWB (Ultra- WideBand) communication. UWB sends a series of rapid pulses, varying their** positions to communicate information.

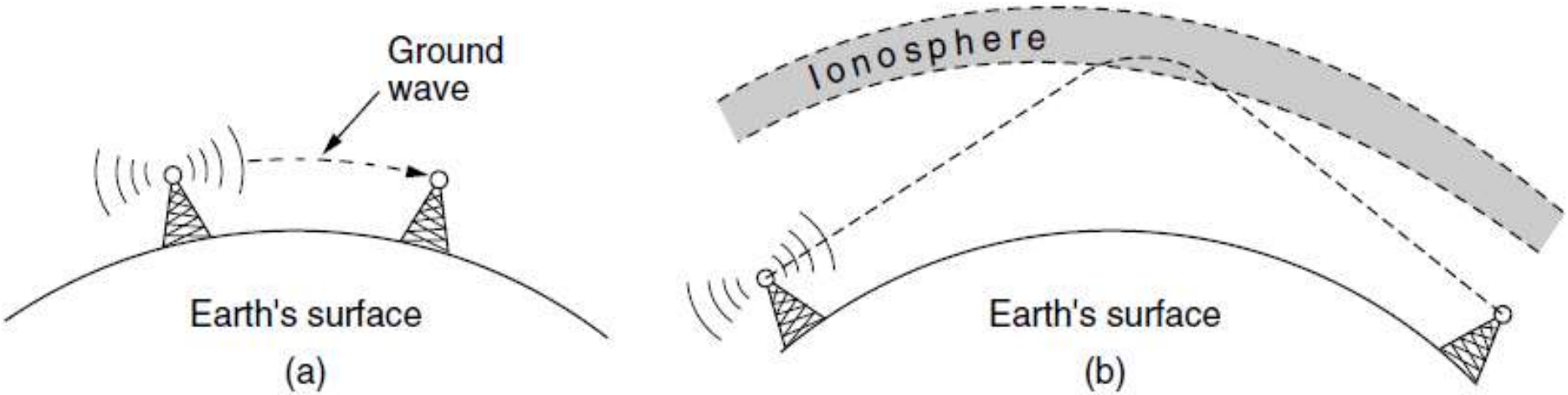
- The rapid transitions lead to a signal that is spread thinly over a very wide frequency band.
- UWB has very little energy at any given frequency when used for short-range transmission, it does not cause harmful interference to those other narrowband radio signals. It is said to **underlay the other signals.**



Spread spectrum and ultra-wideband (UWB) communication.

Radio Transmission

- Radio frequency (RF) waves are easy to generate, can travel long distances, and can penetrate buildings easily, so they are widely used for communication, both indoors and outdoors.
- Radio waves also are omnidirectional, meaning that they travel in all directions from the source, so the transmitter and receiver do not have to be carefully aligned physically.
- The signal energy is spread more thinly over a larger surface. This attenuation is called **path loss.**
- At high frequencies, radio waves tend to travel in straight lines and bounce off obstacles.
- Path loss still reduces power, though the received signal can depend strongly on reflections as well.
- High-frequency radio waves are also absorbed by rain and other obstacles to a larger extent than are low-frequency ones.



(a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth.
(b) In the HF band, they bounce off the ionosphere.

The military also communicate in the HF and VHF bands.

Microwave Transmission

- Microwaves travel in a straight line, so if the towers are too far apart, the earth will get in the way. Thus, repeaters are needed periodically.
- The distance between repeaters goes up very roughly with the square root of the tower height.
- For 100-meter-high towers, repeaters can be 80 km apart.
- Unlike radio waves at lower frequencies, microwaves do not pass through buildings well.
- **Multipath fading** is a serious problem.
- Microwave communication is so widely used for long-distance telephone communication, mobile phones, television distribution, and other purposes that a severe shortage of spectrum has developed.
- Microwave is also relatively inexpensive.

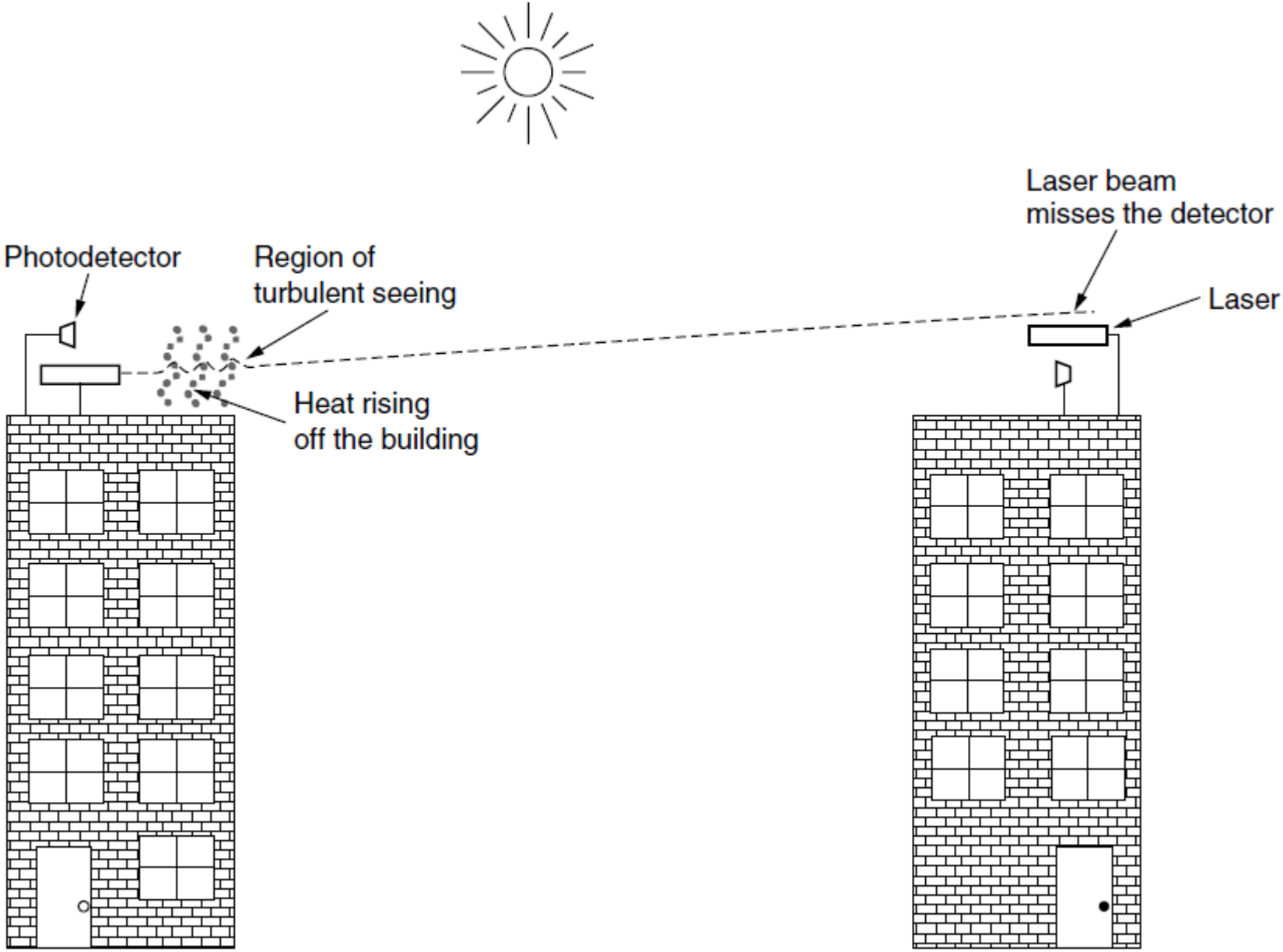
- Putting up two simple towers (which can be just big poles with four guy wires) and putting antennas on each one may be cheaper than burying 50 km of fiber through a congested urban area or up over a mountain, and it may also be cheaper than leasing the telephone company's fiber, especially if the telephone company has not yet even fully paid for the copper it ripped out when it put in the fiber.

Infrared Transmission

- Widely used for short-range communication.
- The remote controls used for televisions, VCRs, and stereos all use infrared communication.
- They are relatively directional, cheap, and easy to build.
- A major drawback is they do not pass through solid objects.
- Infrared communication has a limited use on the desktop, for example, to connect notebook computers and printers with the **IrDA (Infrared Data Association) standard,** but it is not a major player in the communication game.

Light Transmission

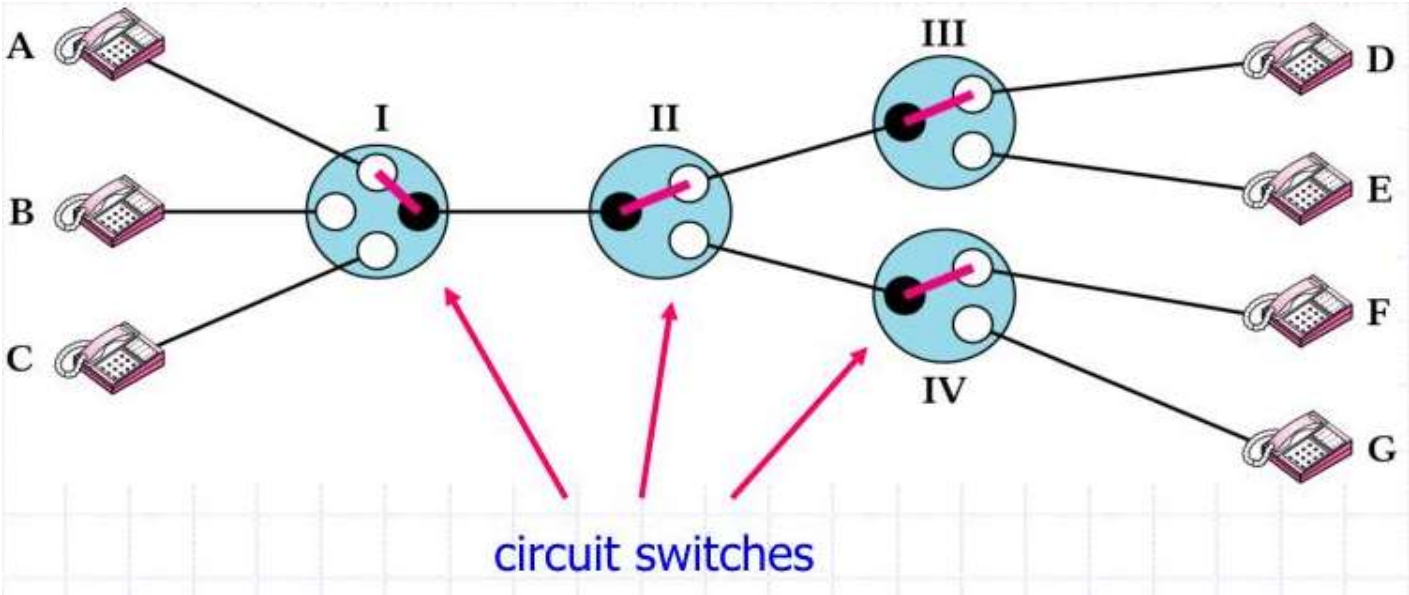
- A more modern application is to connect the LANs in two buildings via lasers mounted on their rooftops.
- This scheme offers very high bandwidth at very low cost and is relatively secure because it is difficult to tap a narrow laser beam. It is also relatively easy to install and, unlike microwave transmission, does not require an FCC license.



Convection currents can interfere with laser communication systems. A bidirectional system with two lasers is pictured here.

Circuit switching

- Circuit switching is a communication method where a dedicated communication path, or circuit, is established between two devices before data transmission begins.
- The circuit remains dedicated to the communication for the duration of the session, and no other devices can use it while the session is in progress.
- Circuit switching is commonly used in voice communication and some types of data communication.



Advantages of Circuit Switching:

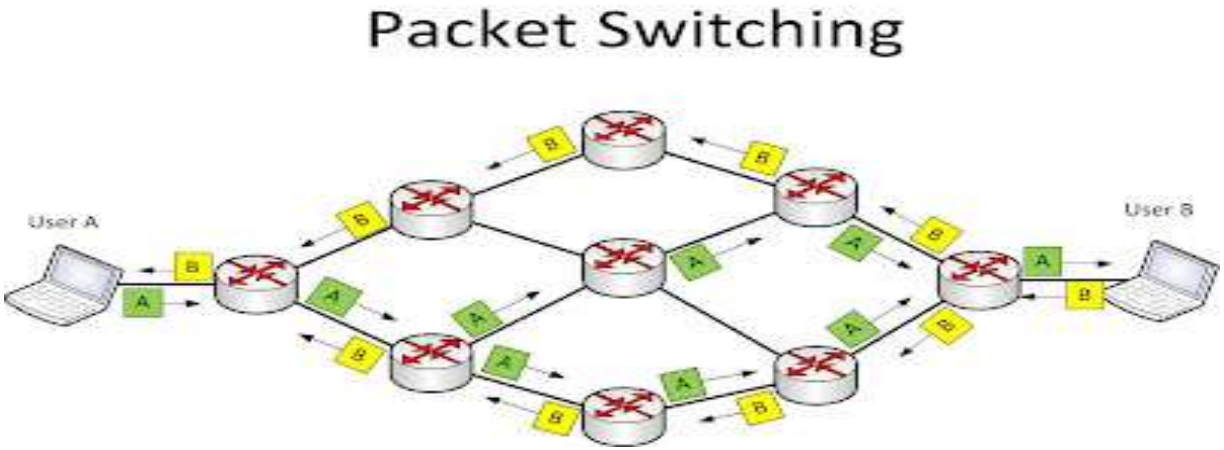
- **Guaranteed bandwidth:** Circuit switching provides a dedicated path for communication, ensuring that bandwidth is guaranteed for the duration of the call.
- **Low latency:** Circuit switching provides low latency because the path is predetermined, and there is no need to establish a connection for each packet.
- **Predictable performance:** Circuit switching provides predictable performance because the bandwidth is reserved, and there is no competition for resources.
- **Suitable for real-time communication:** Circuit switching is suitable for real-time communication, such as voice and video, because it provides low latency and predictable performance.

Disadvantages of Circuit Switching:

- **Inefficient use of bandwidth:** Circuit switching is inefficient because the bandwidth is reserved for the entire duration of the call, even when no data is being transmitted.
- **Limited scalability:** Circuit switching is limited in its scalability because the number of circuits that can be established is finite, which can limit the number of simultaneous calls that can be made.
- **High cost:** Circuit switching is expensive because it requires dedicated resources, such as hardware and bandwidth, for the duration of the call.

Packet switching

- Packet switching is a communication method where **data is divided into smaller units** called **packets** and transmitted over the network.
- Each packet contains the **source and destination** addresses, as well as other information needed for routing.
- The packets may take different paths to reach their destination, and they may be transmitted out of order or delayed due to network congestion.



Advantages of Packet Switching:

- **Efficient use of bandwidth:** Packet switching is efficient because bandwidth is shared among multiple users, and resources are allocated only when data needs to be transmitted.
- **Flexible:** Packet switching is flexible and can handle a wide range of data rates and packet sizes.
- **Scalable:** Packet switching is highly scalable and can handle large amounts of traffic on a network.
- **Lower cost:** Packet switching is less expensive than circuit switching because resources are shared among multiple users.

Disadvantages of Packet Switching:

- **Higher latency:** Packet switching has higher latency than circuit switching because packets must be routed through multiple nodes, which can cause delay.
- **Limited QoS:** Packet switching provides limited QoS guarantees, meaning that different types of traffic may be treated equally.
- **Packet loss:** Packet switching can result in packet loss due to congestion on the network or errors in transmission.
- **Unsuitable for real-time communication:** Packet switching is not suitable for real-time communication, such as voice and video, because of the potential for latency and packet loss.